



北京大学
PEKING UNIVERSITY

Novel Ferroelectric Tunnel FinFET based Encryption-embedded Computing-in-Memory for Secure AI with High Area- and Energy-Efficiency

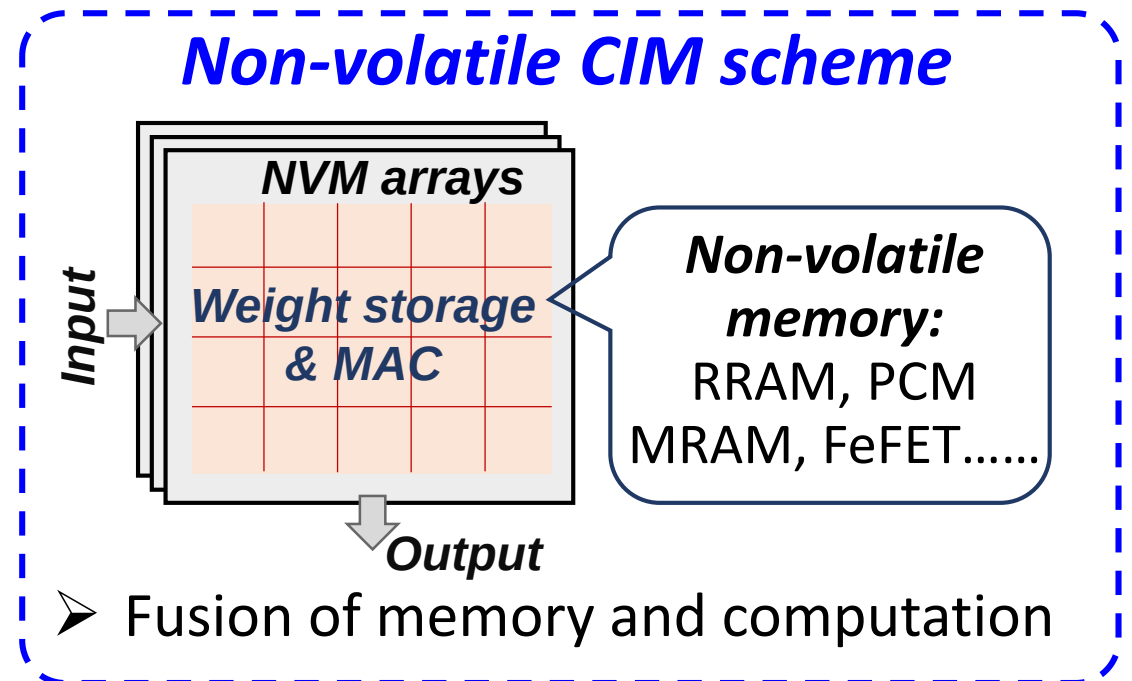
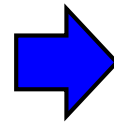
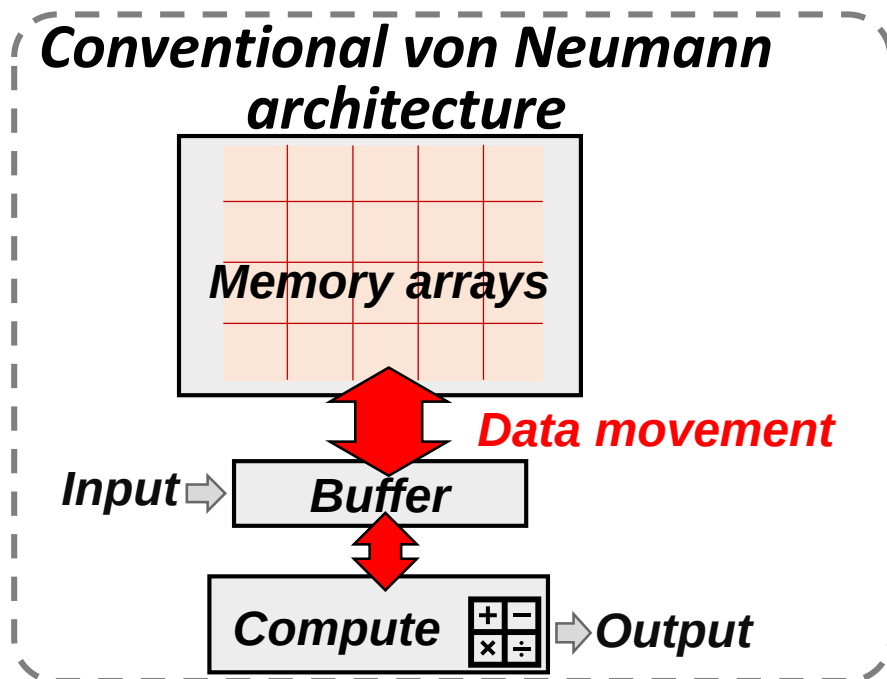
Jin Luo, Hanyong Shao, Boyi Fu, Zhiyuan Fu, Weikai Xu, Kaifeng Wang,
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Motivation (1/4)

Non-volatile computing-in-memory (nvCIM)

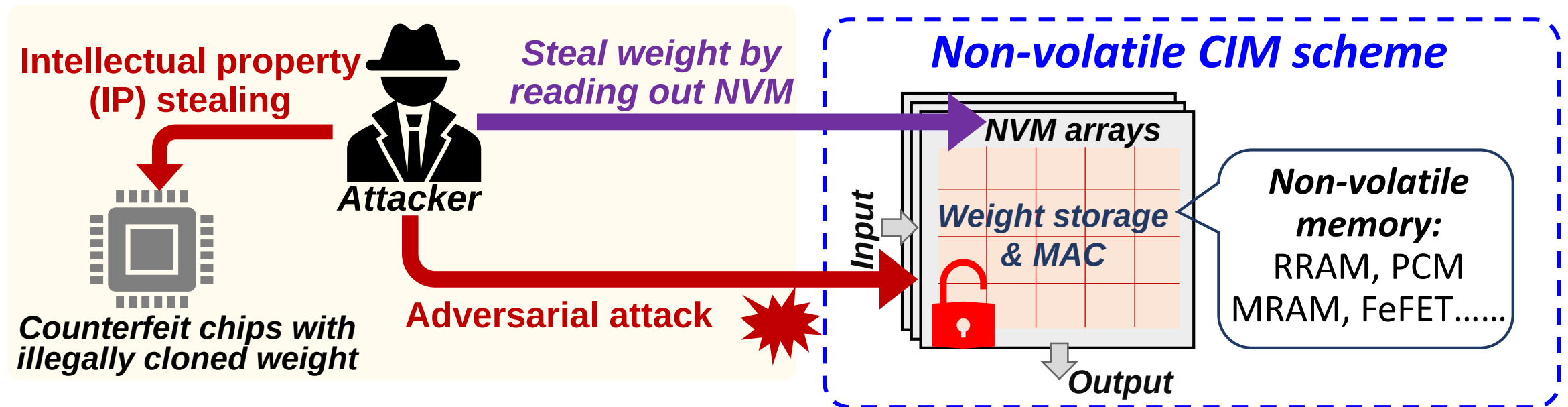
- The conventional Von Neumann architecture: **memory wall** bottleneck
- Non-volatile memory (NVM) based computing-in-memory (nvCIM) scheme
 - Store weight of artificial neural network (ANN) with **NVM** device
 - **Highly-parallel multiply-accumulate (MAC)** inside NVM arrays



Motivation (2/4)

Security vulnerabilities of nvCIM

- Security vulnerabilities of nvCIM based on NVM array
 - The **raw weight (W)** parameters of ANN are **stored in NVM** and are **easy to be retrieved** by directly physical accessing the NVM
 - **Intellectual property (IP) stealing** and **adversarial attack**

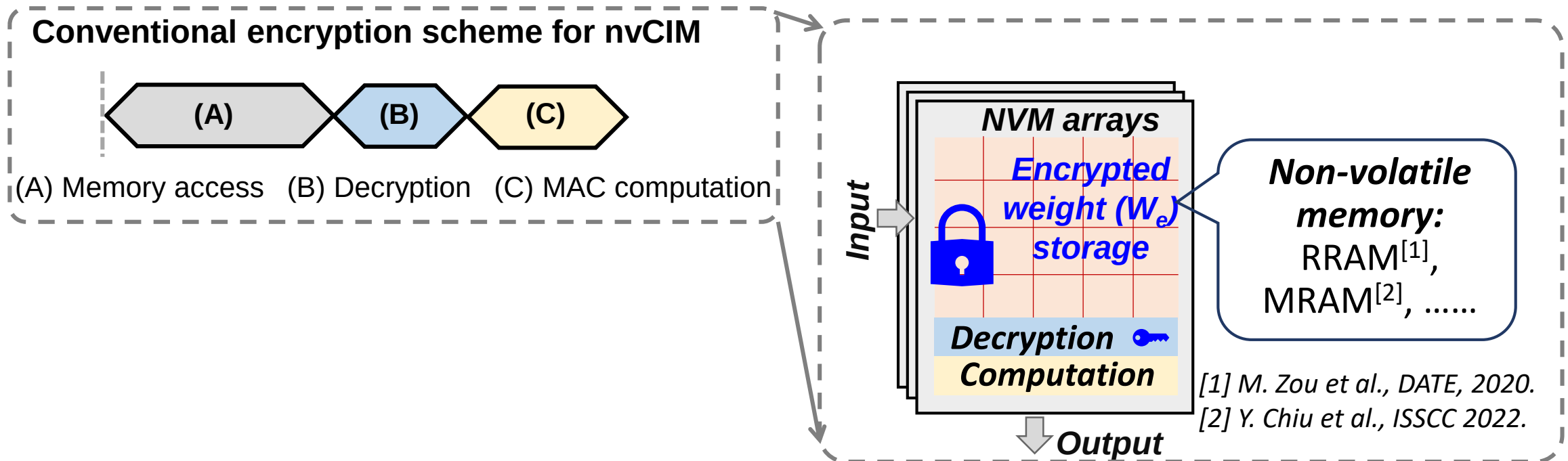


Encrypting the NVM-based CIM is of great importance for secure AI

Motivation (3/4)

Conventional encryption scheme for nvCIM

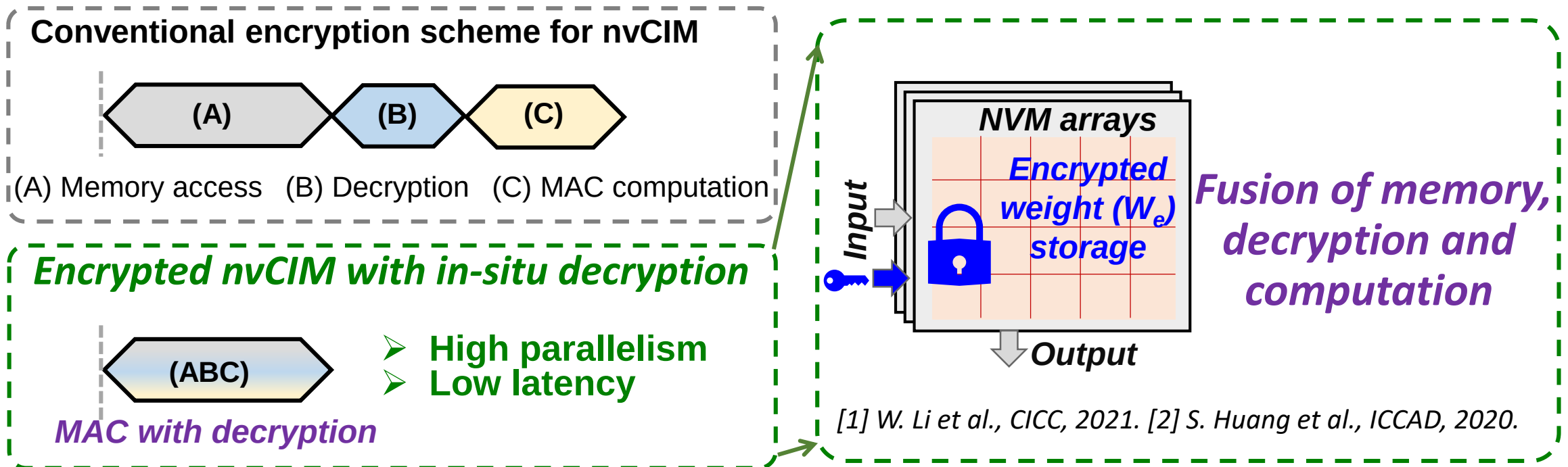
- Hardware implementation of the encryption for NVM array based nvCIM
 - **Storage and readout of encrypted weight (W_e)** in NVM
 - Adjacently **extra decryption circuit** for decrypting W_e
 - **Peripheral digital unit** for executing computation
- Sacrificing parallelizability & introducing extra circuit overhead*



Motivation (4/4)

Encrypted nvCIM with in-situ decryption

- **In-situ decryption** along with parallel MAC inside NVM array
 - Merge the authorized *key* for W_e decryption into the *input vector* (X)
 - Perform **high parallel analog computation** in the W_e array with implicit decryption
 - **Eliminating explicit decryption** process and extra circuit



In this work

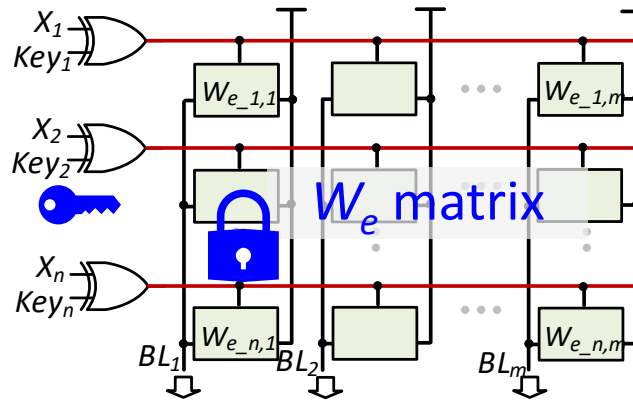
- Implement **encrypted nvCIM** by ***in-situ* decryption** with high area- & energy-efficiency
 - Algorithmically modify encryption with X(N)OR operation, enabling encrypted MAC in W_e
 - Implement W_e of XNOR operation with only **one ferroelectric tunnel FET (FeTFET)**

Novel encrypted MAC (eMAC) algorithm

Intrinsic associative law of XOR-cipher and XNOR-MAC



In-situ eMAC in W_e cell of XNOR operation



In this work

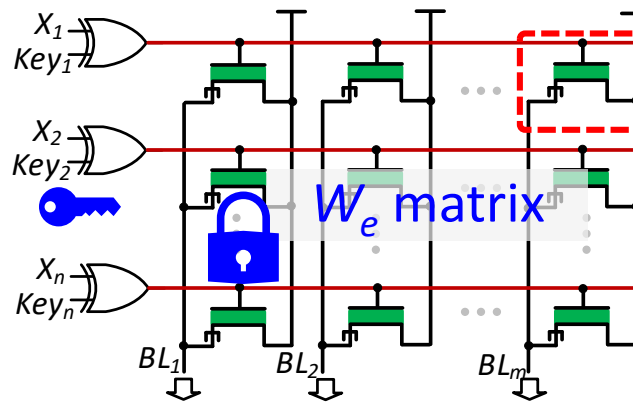
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Novel encrypted MAC (eMAC) algorithm

Intrinsic associative law of XOR-cipher and XNOR-MAC



In-situ eMAC in W_e cell of XNOR operation



W_e cell of XNOR operation with only 1 FeTFET

FE spontaneous polarization



Non-volatile W_e storage

Three-terminal structure with gated ambipolar band-to-band tunneling



XNOR operation of in-situ eMAC within W_e

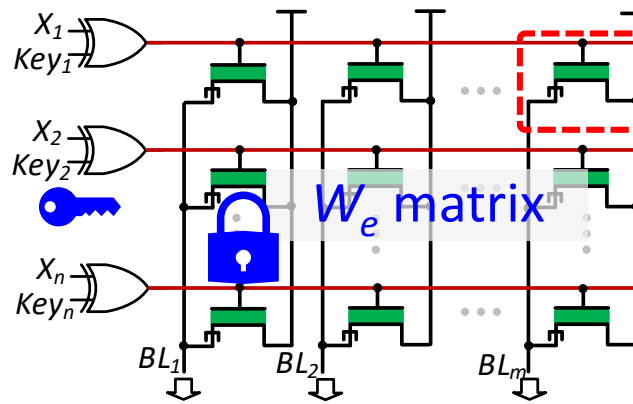
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XNOR operation of in-situ eMAC within W_e

FeTFET-based encrypted nvCIM for secure AI

- ✓ Encrypted inference and learning process of ANNs
- ✓ **High area- and energy-efficiency**

Outline

- Design of FeTFET-based encrypted MAC (eMAC)
- Experiments of FeTFET-based eMAC behavior
- FeTFET-based encrypted nvCIM macro for edge secure AI
- Summary

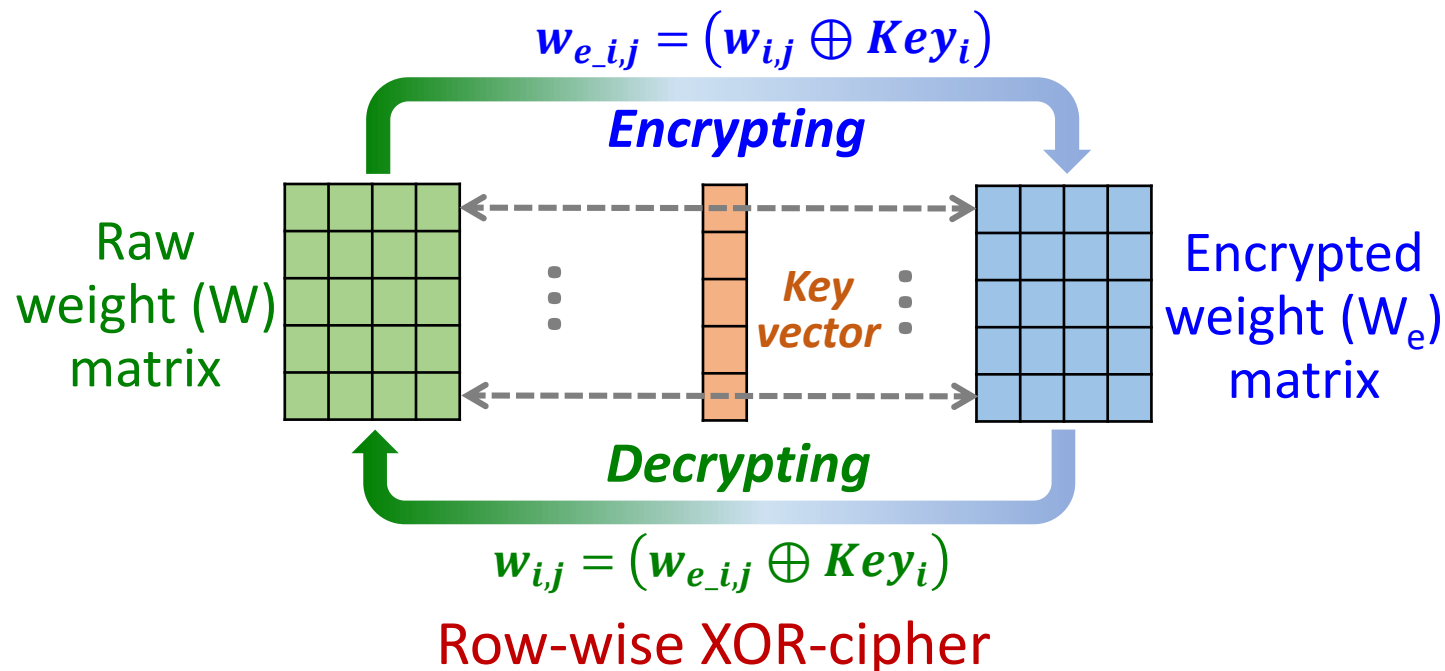
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Encrypted MAC (eMAC) (1/3)

—XOR-cipher for weight encryption

- Row-wise XOR-cipher for encryption of weight matrix in ANNs
 - XOR-cipher: an effective symmetric cryptography method for weight encryption
 - Encrypted weight (W_e) matrix is obtained through row-wise XOR operation between raw weight (W) matrix and authorized *key* bit-vector, and vice versa.
 - W_e matrix will be stored in memory array



Logic truth table of XOR-cipher

W	Key	W_e
0	0	0
1	0	1
0	1	1
1	1	0

Encrypted MAC (eMAC) (2/3)

—*In-situ* parallel encrypted MAC

- Implicitly merging decryption process of XOR-cipher into MAC operation
 - Previous work: utilizing AND-based multiplication^{[1]-[2]}
 - Binomial operation in W_e cell, requiring **complementary form of W_e storage and Key input**

AND-based MAC with XOR-cipher

$$y_j = \sum_i^n x_i \cdot w_{i,j}$$

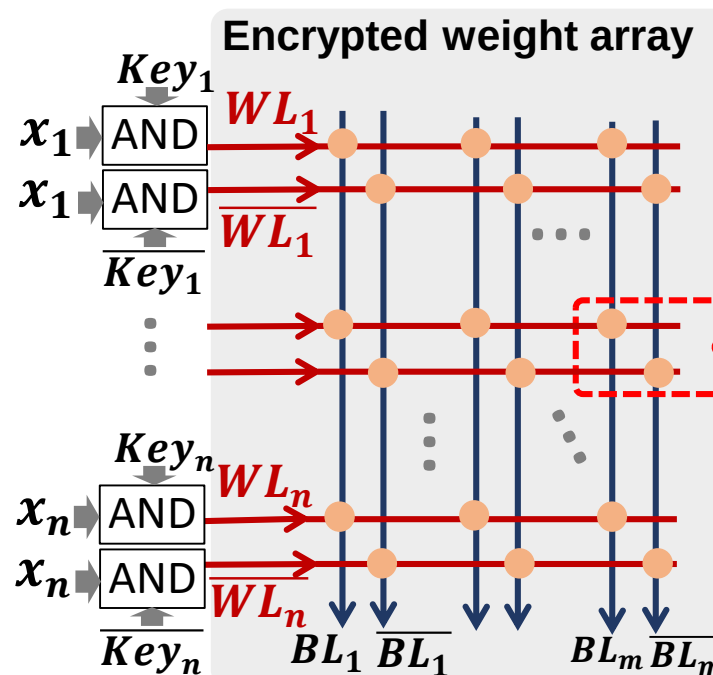
$$= \sum_i^n x_i \cdot (w_{eij} \oplus Key_i)$$

$$= \sum_i^n x_i \cdot (\overline{w_{eij}} \cdot Key_i + w_{eij} \cdot \overline{Key_i})$$

$$= \sum_i^n (x_i \cdot Key_i) \cdot \overline{w_{eij}} + \sum_i^n (x_i \cdot \overline{Key_i}) \cdot w_{eij}$$

Binomial operation in each W_e cell

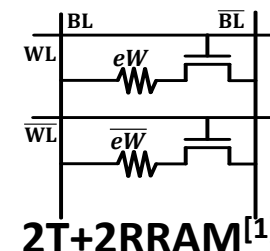
X	"0"	"0"	"1"	"1"
W	"0"	"1"	"0"	"1"
X · W	"1"	"0"	"0"	"1"



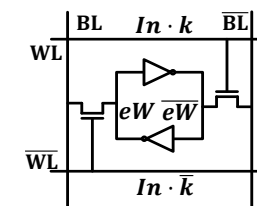
WL: word-line

BL: Bit-line

Complementary W_e cell



2T+2RRAM^[1]



SRAM with split WL^[2]

Encrypted MAC (eMAC) (3/3)

—*In-situ* parallel encrypted MAC

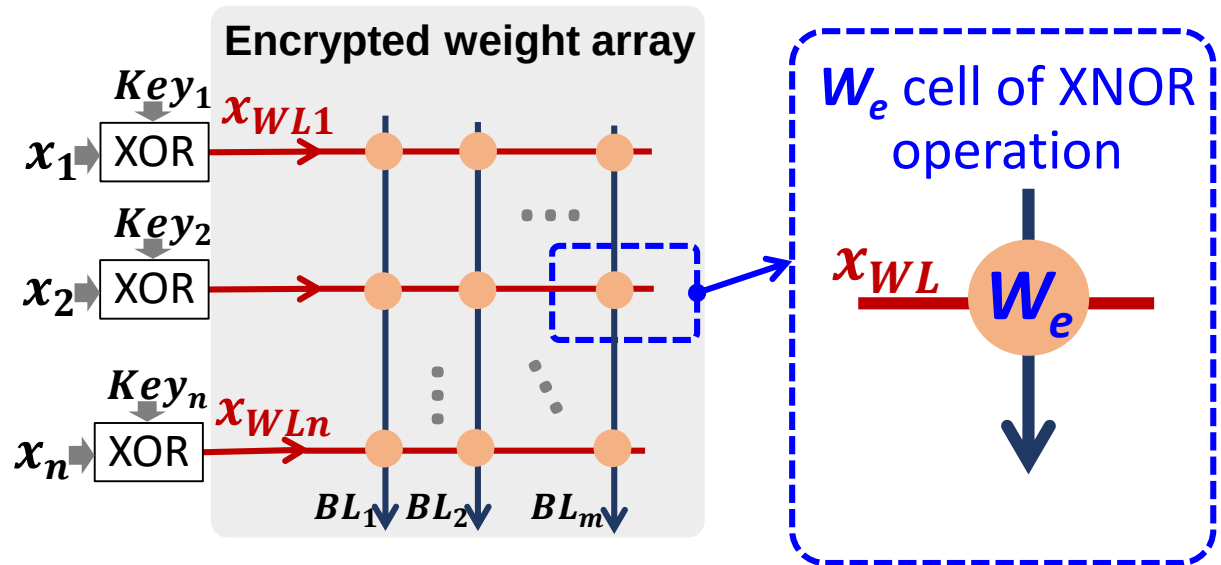
- Implicitly merging decryption process of XOR-cipher into MAC operation
 - **This work:** utilizing XNOR-based multiplication
 - Monomial operation of XNOR in W_e cell, by taking advantage of associative law of X(N)OR

XNOR-based MAC with XOR-cipher

$$\begin{aligned}
 y_j &= \sum_i^n x_i \odot w_{i,j} \\
 &= \sum_i^n x_i \odot (w_{e_{i,j}} \oplus \text{Key}_i) \\
 &= \sum_i^n \underbrace{(x_i \oplus \text{Key}_i)}_{x_{WL,i}} \odot w_{e_{i,j}}
 \end{aligned}$$

X	“-1”	“-1”	“+1”	“+1”
W	“-1”	“+1”	“-1”	“+1”
$X \cdot W$	“+1”	“-1”	“-1”	“+1”

Monomial operation of XNOR in each W_e cell



Encrypted MAC (eMAC) (3/3)

—*In-situ* parallel encrypted MAC

- Implicitly merging decryption process of XOR-cipher into MAC operation
 - **This work:** utilizing XNOR-based multiplication
 - W_e cell enabling XNOR operation between X_{WL} and W_e can be implemented with one FeTFET
 - Without the need of complementary form of W_e storage, key input, WLs and BLs.

XNOR-based MAC with XOR-cipher

$$y_j = \sum_i^n x_i \odot w_{i,j}$$

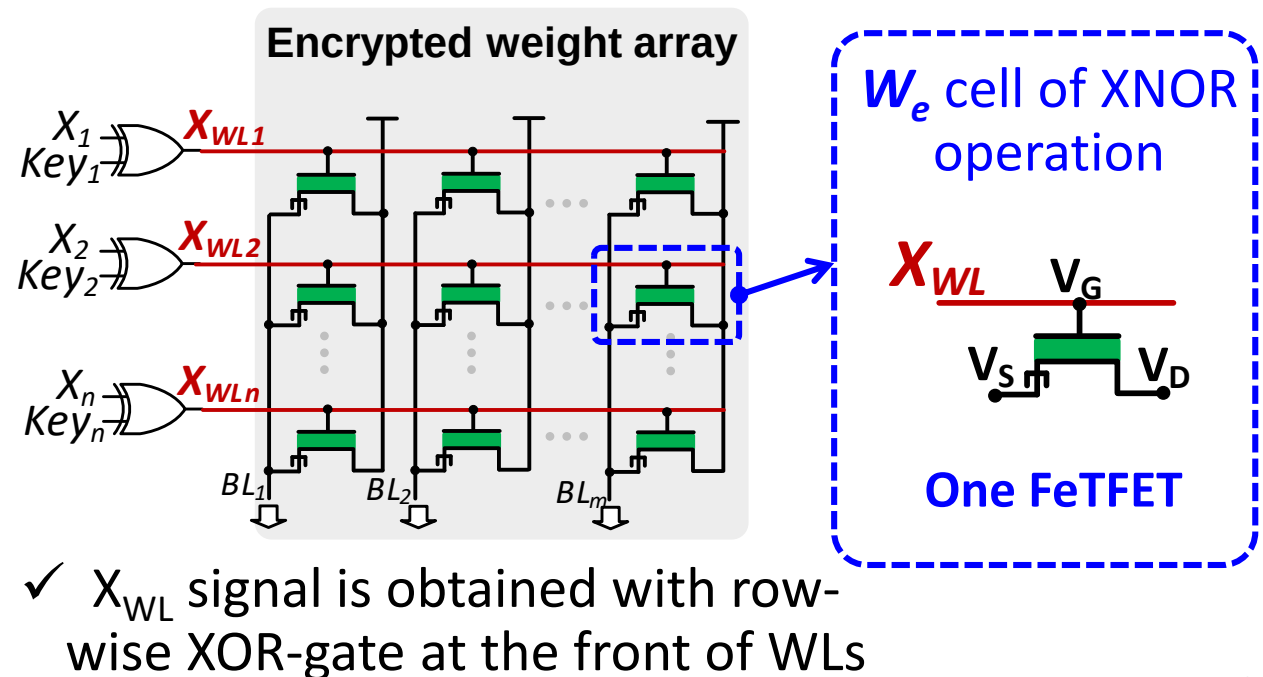
$$= \sum_i^n x_i \odot (w_{e_{i,j}} \oplus Key_i)$$

$$= \sum_i^n (x_i \oplus Key_i) \odot w_{e_{i,j}}$$

$x_{WL,i}$

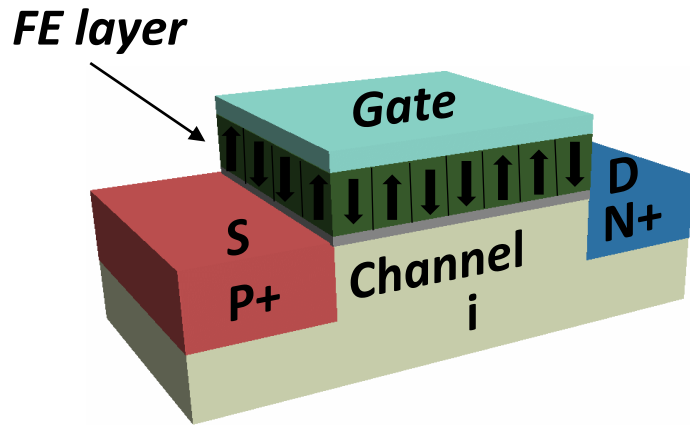
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W	“-1”	“+1”	“-1”	“+1”
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Monomial operation of XNOR in each W_e cell

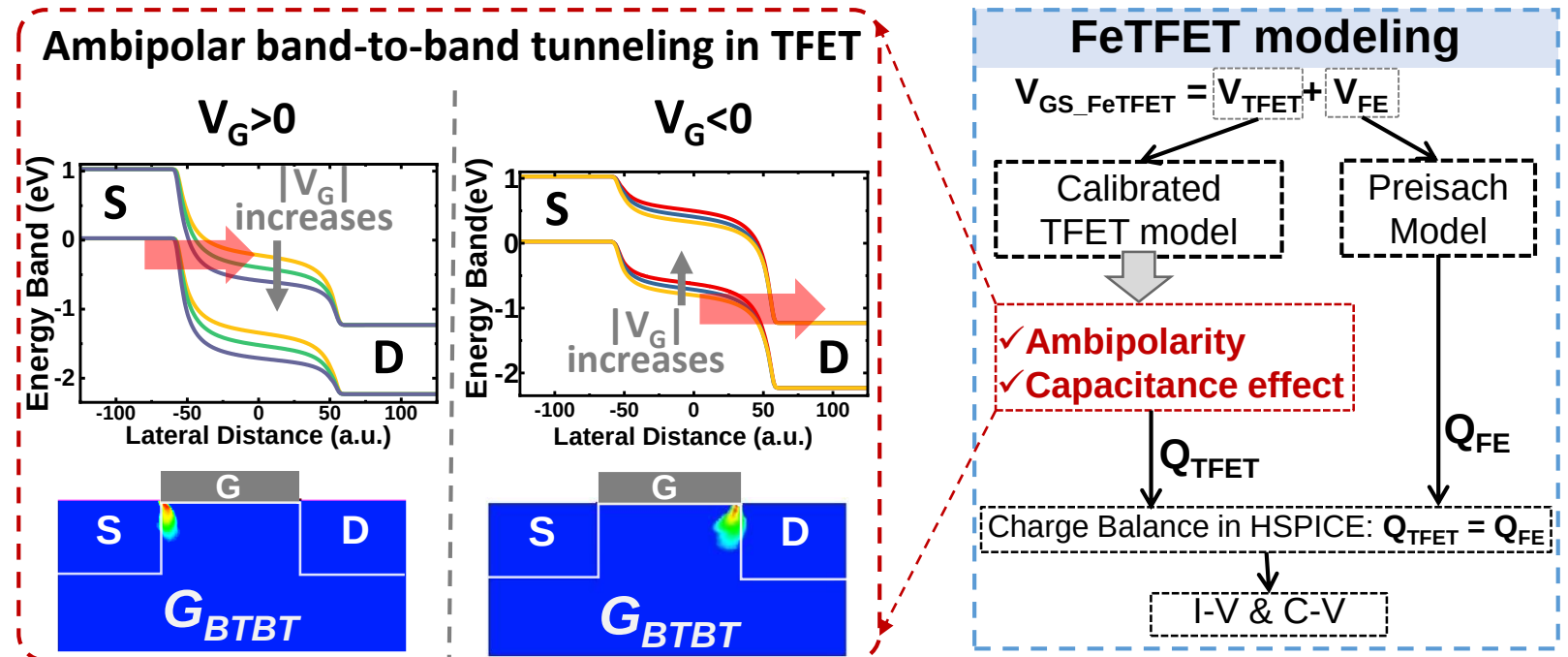


FeTFET mechanism for XNOR operation (1/3)

- Device structure of FeTFET
 - Ferroelectric (FE) layer integrated in the gate stack of tunnel FET (TFET)
- Ambipolar transport mechanism of conventional TFET
 - Band-to-band tunneling (BTBT) in source junction ($V_G > 0$) or drain junction ($V_G < 0$)

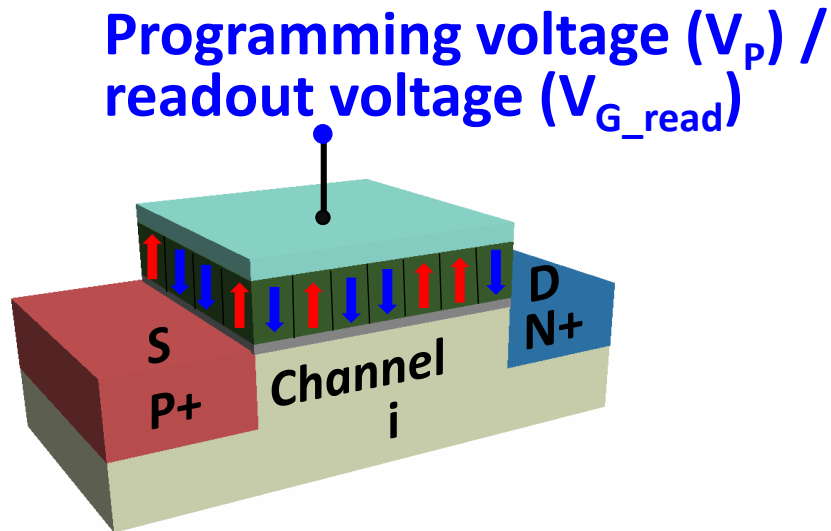


FeTFET structure

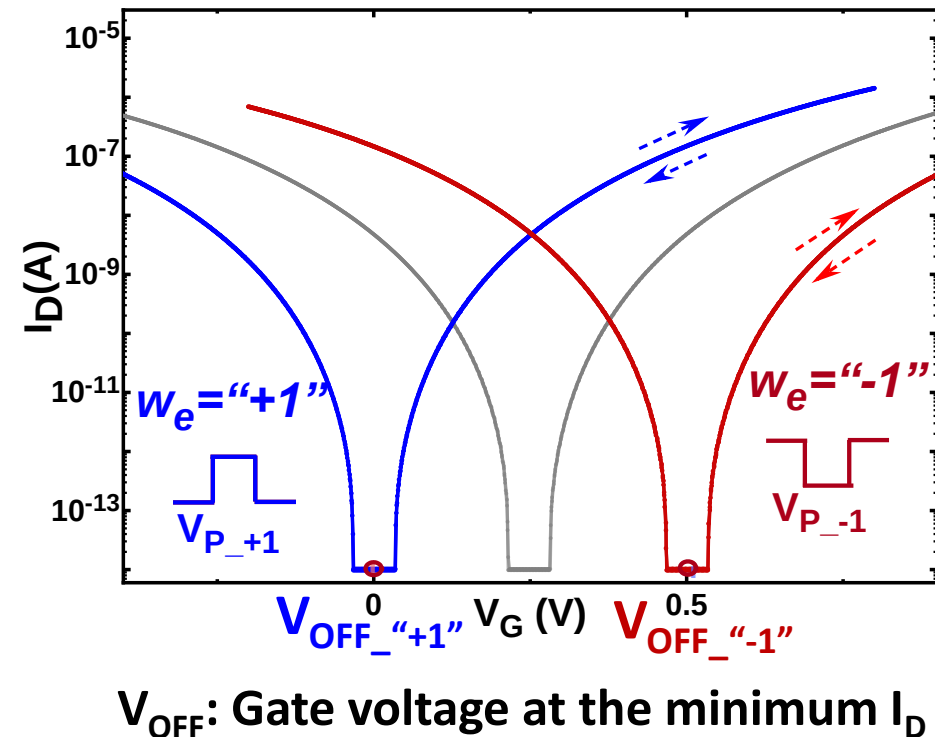


FeTFET mechanism for XNOR operation (2/3)

- Non-volatile FE polarization for W_e storage with programming voltage (V_p)
 - Modulate threshold voltage (V_{TH}) $\rightarrow$ Modulation of V_{OFF} for W_e storage with “-1”/“+1” (V_{OFF} : gate-voltage corresponding to the minimum I_D)
 - Gated ambipolar BTBT current I_D in non-destructive readout mode w/o FE switching

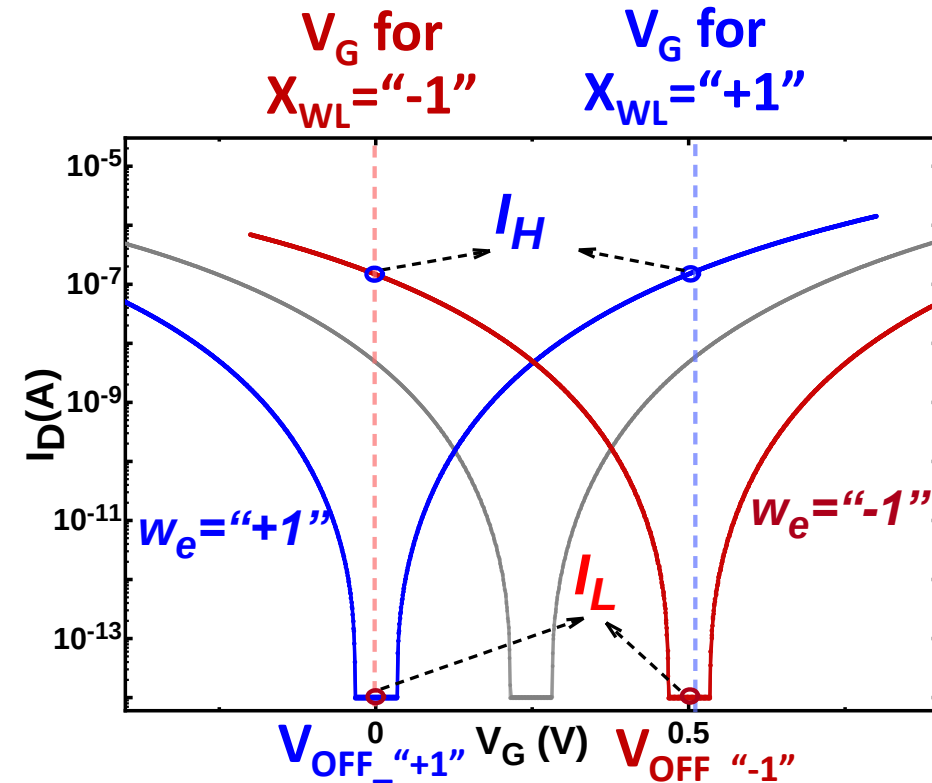
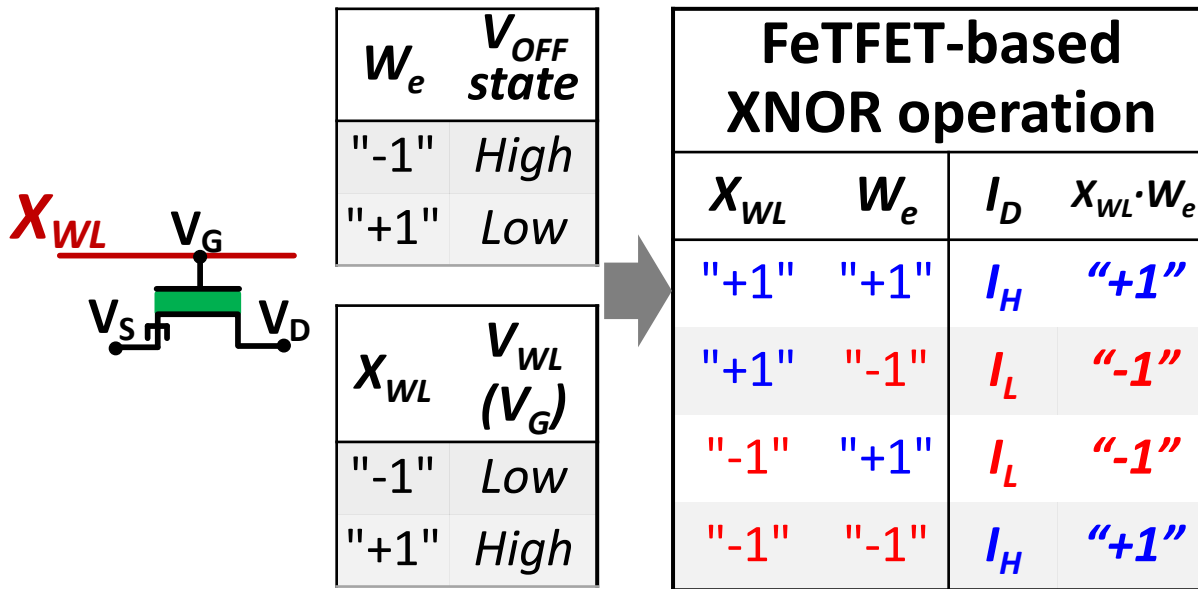


- ← Forward FE polarization
→ Reverse FE polarization



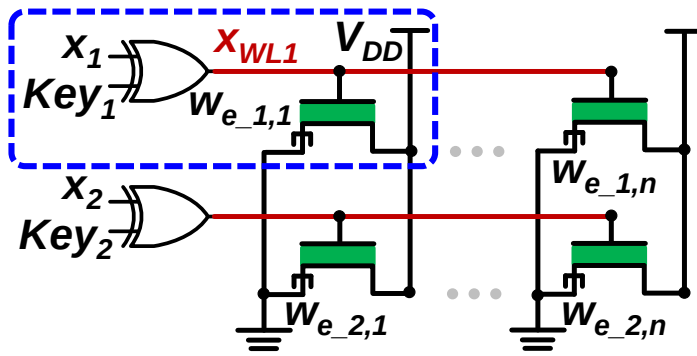
FeTFET mechanism for XNOR operation (3/3)

- XNOR operation^[1] of W_e and X_{WL} in **one FeTFET**
 - The distinct I_D difference represents the results of “-1”/“+1” for XNOR-based signed multiplication between X_{WL} and W_e .

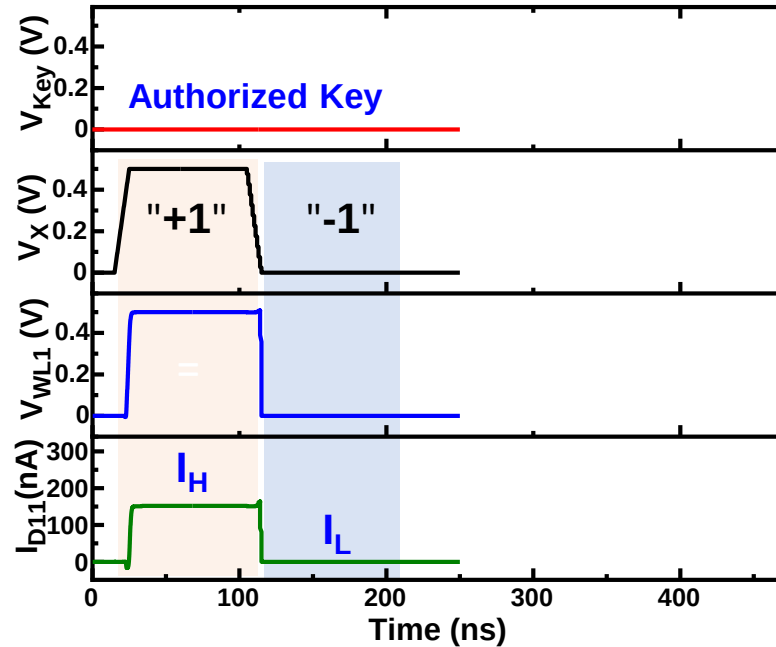


FeTFET-based eMAC array analysis (1/3)

- Encrypted multiplication of W and X inside one FeTFET based W_e cell
 - With authorized key: proper XNOR-based multiplication without explicit decryption



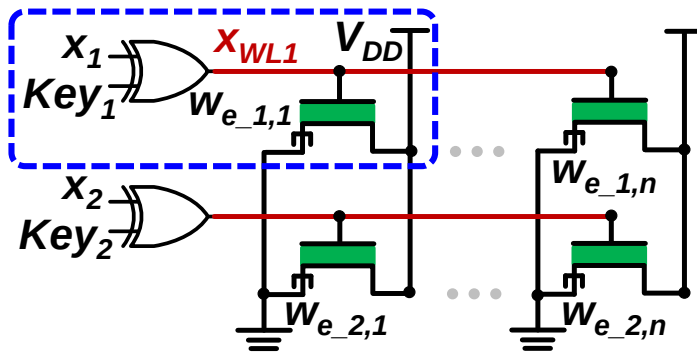
- $W_{e,1,1}$ = "+1" when $W_{1,1}$ = "+1" & Key=0



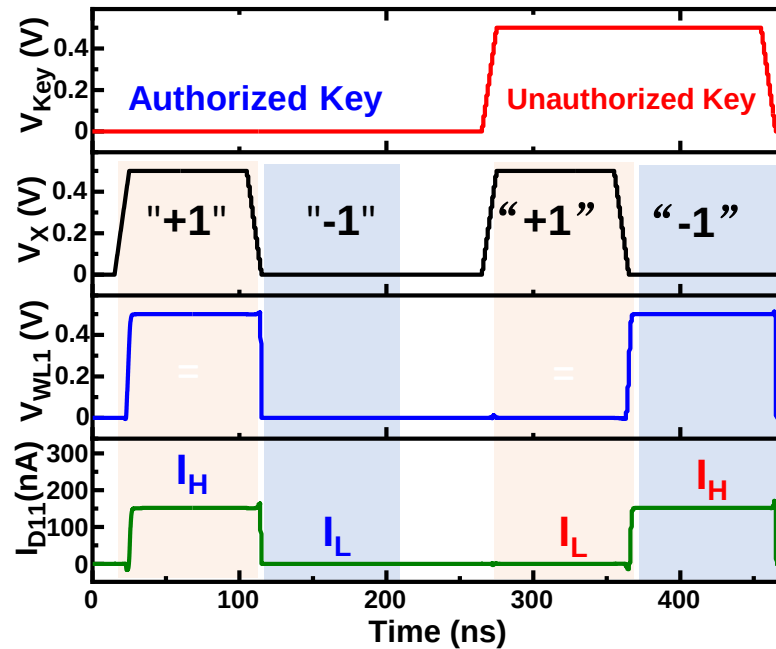
FeTFET-based encrypted signed multiplication				
X	W	Authorized Key?	$(X \oplus \text{Key}) \odot w_e$	I_D
"+1"	"+1"	✓	$X \odot W$	I_H
"+1"	"-1"	✓	$X \odot W$	I_L
"-1"	"+1"	✓	$X \odot W$	I_L
"-1"	"-1"	✓	$X \odot W$	I_H

FeTFET-based eMAC array analysis (2/3)

- Encrypted multiplication of W and X inside one FeTFET based W_e cell
 - With authorized key: proper XNOR-based multiplication without explicit decryption
 - With unauthorized wrong key: wrong multiplication opposite in sign



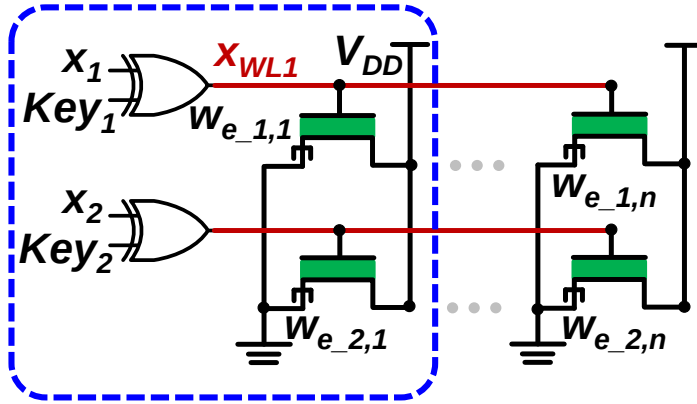
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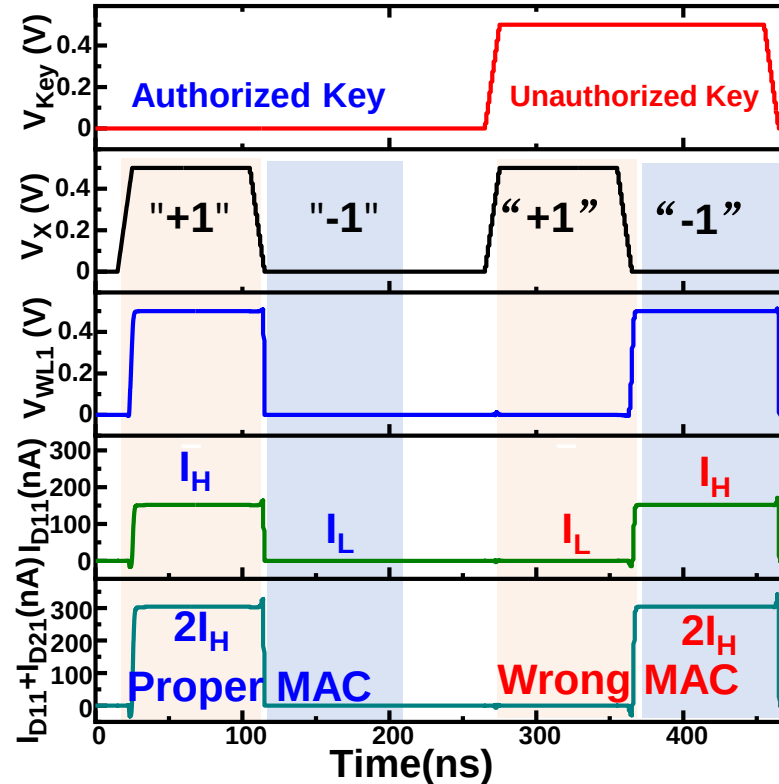
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" +1 "	" -1 "	✓	$X \odot W$	I_L
" -1 "	" +1 "	✓	$X \odot W$	I_L
" -1 "	" -1 "	✓	$X \odot W$	I_H
" +1 "	" +1 "	✗	$\overline{X \odot W}$	I_L
" +1 "	" -1 "	✗	$\overline{X \odot W}$	I_H
" -1 "	" +1 "	✗	$\overline{X \odot W}$	I_H
" -1 "	" -1 "	✗	$\overline{X \odot W}$	I_L

FeTFET-based eMAC array analysis (3/3)

- Proper MAC results are obtained through bit-line current (I_{BL}) of FeTFET-based W_e array only when using authorized key



- $W_{e_1,1}$ = "+1" when W_1 = "+1" & Key=0
- $w_{1,1} = w_{2,1}$; $Key_1 = Key_2$; $x_1 = x_2$



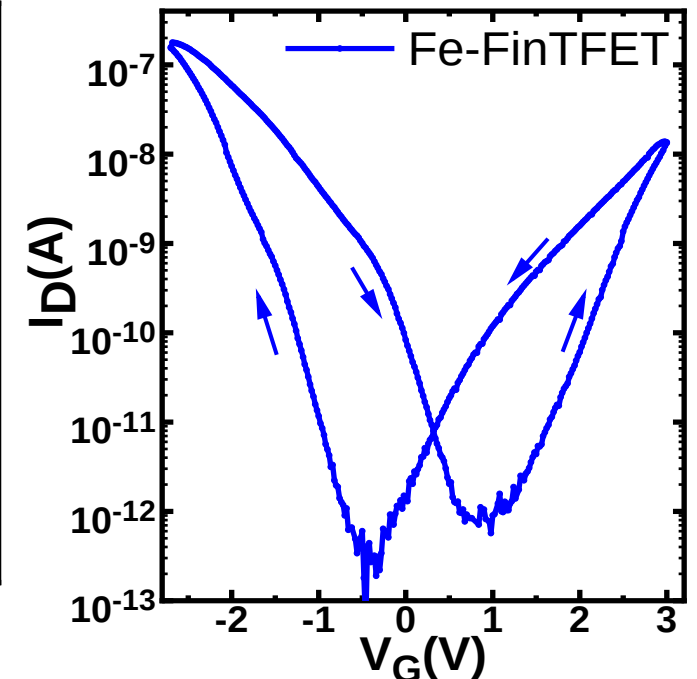
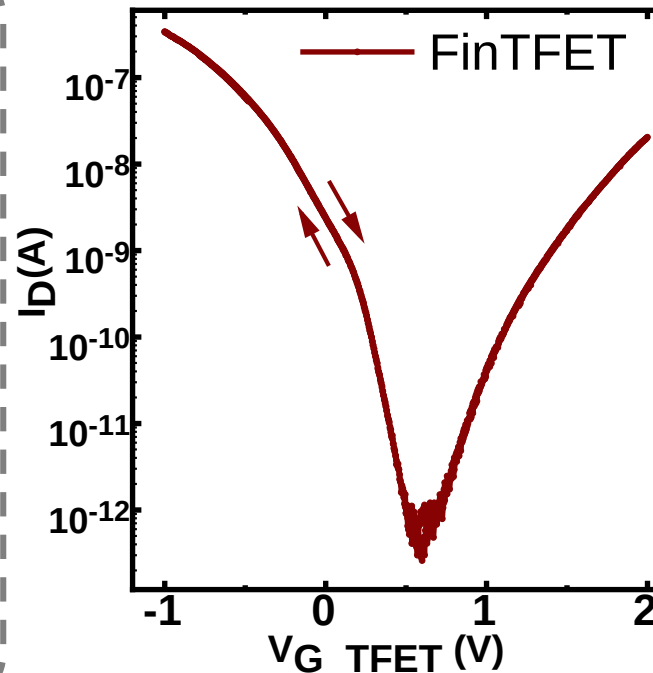
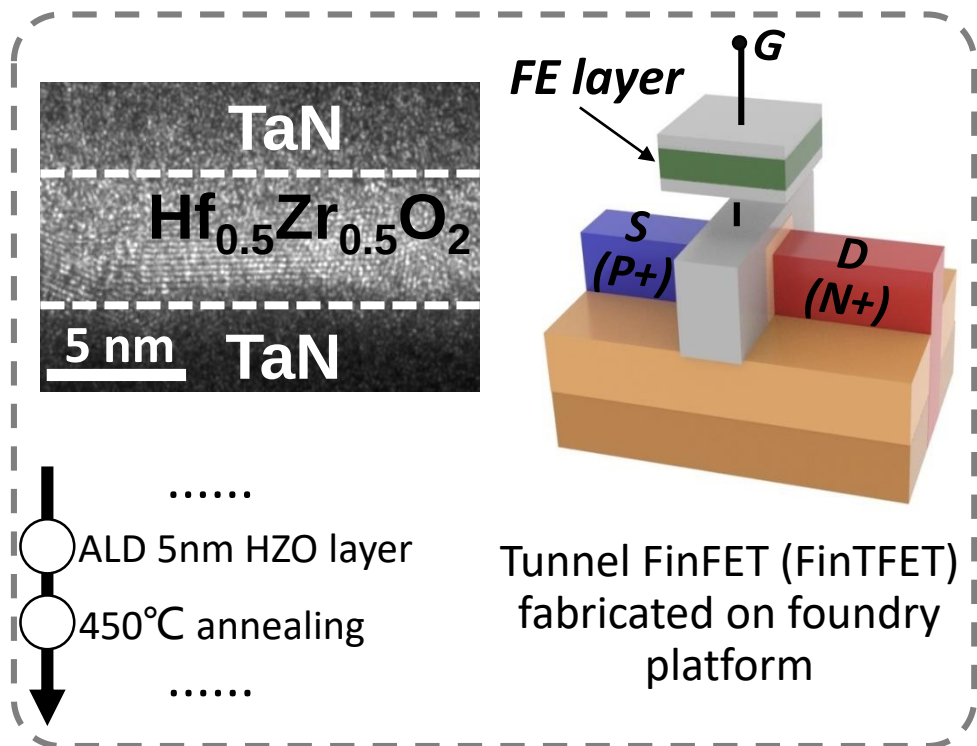
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" -1 "	" -1 "	✗	$\overline{X \odot W}$	I_L

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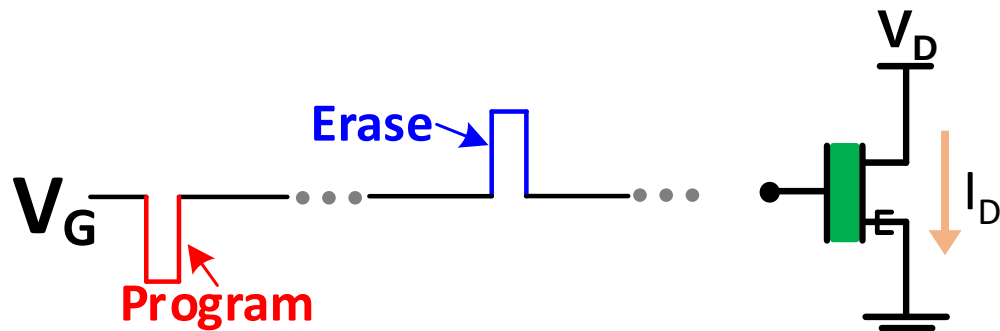
Characteristics of Fe-FinTFET (1/4)

- The experimental FeTFET is fabricated based on FinTFET structure
 - The designed FinTFET without drain underlap region shows ambipolar I_D - V_G .
 - I_D - V_G curve of Fe-FinTFET : typical FE hysteresis with ambipolarity

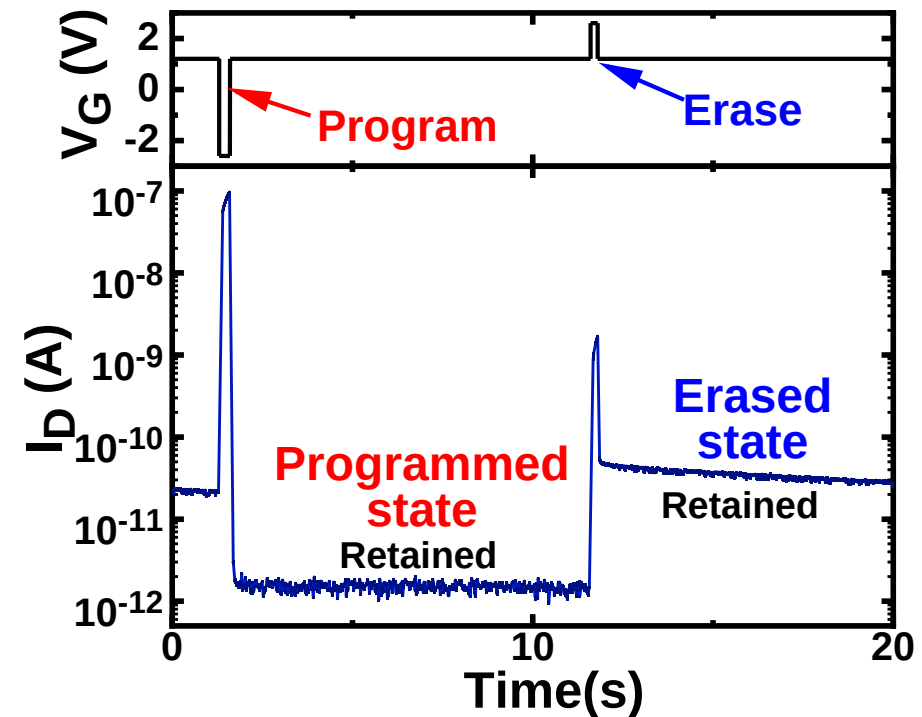


Characteristics of Fe-FinTFET (2/4)

- Program/erase mode with large gate voltage of $|V_p|$
 - Non-volatile ferroelectric (FE) polarization behavior, indicating W_e storage

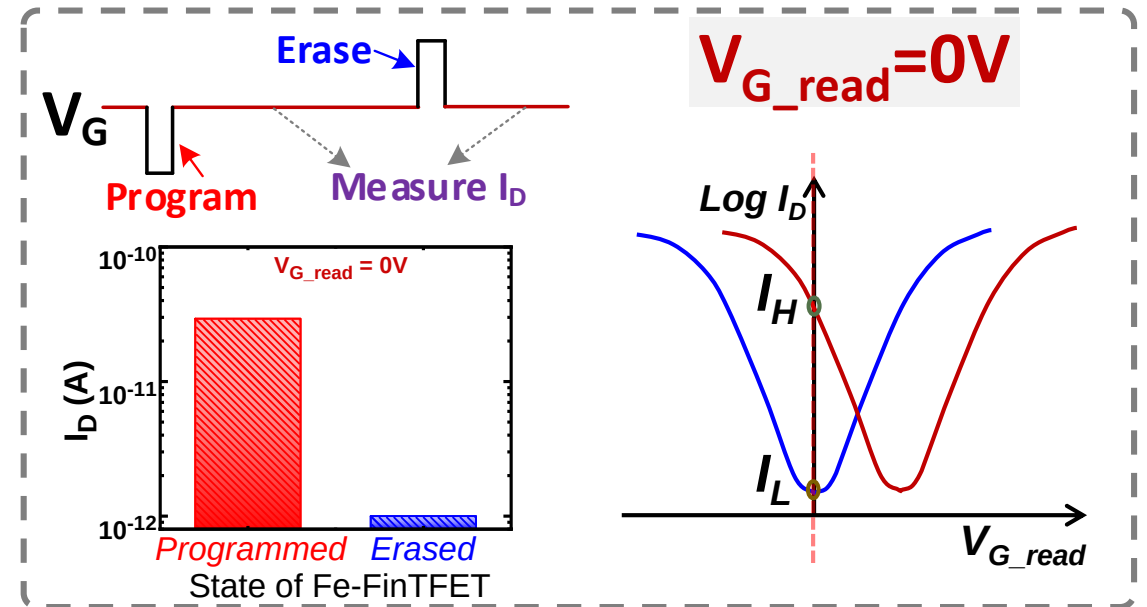
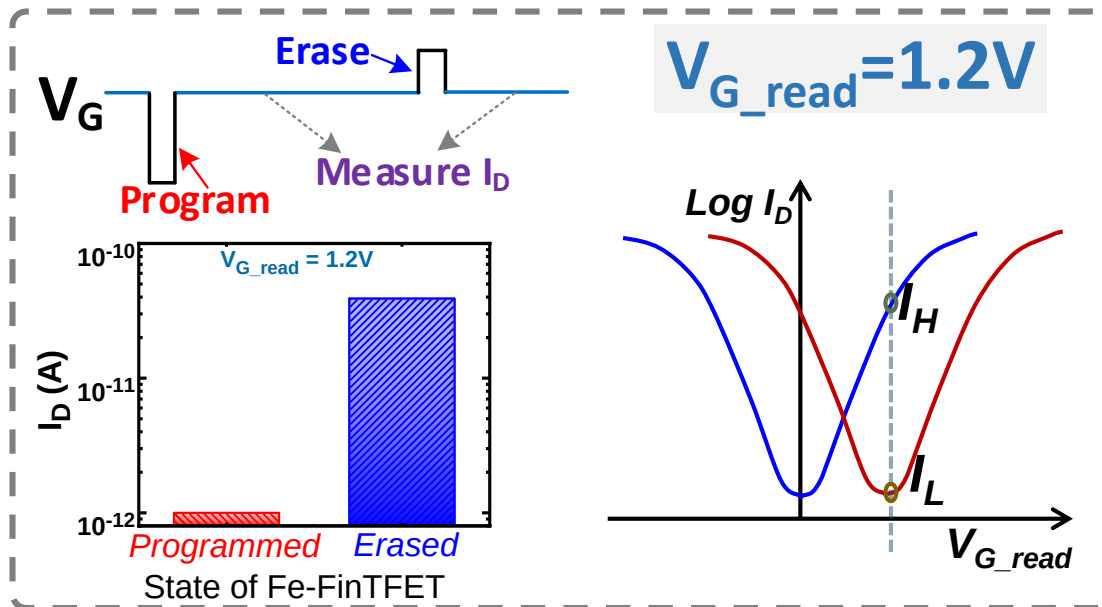


Monitor the I_D over time with programming/erasing pulse applied



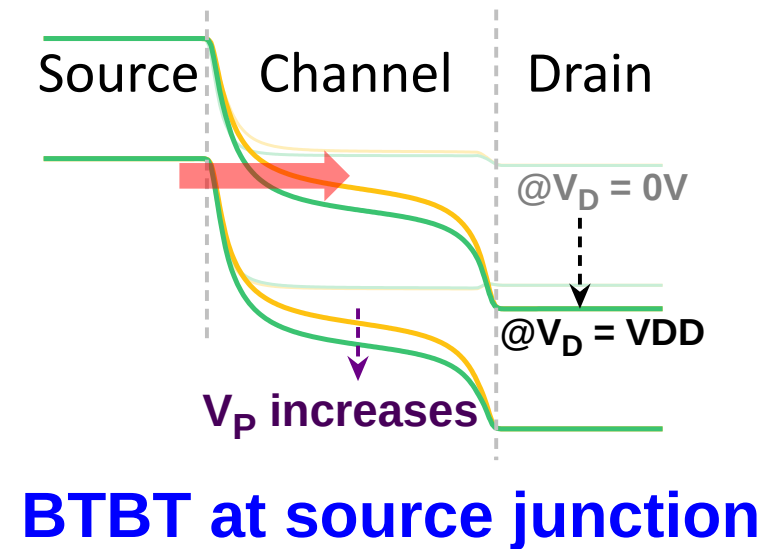
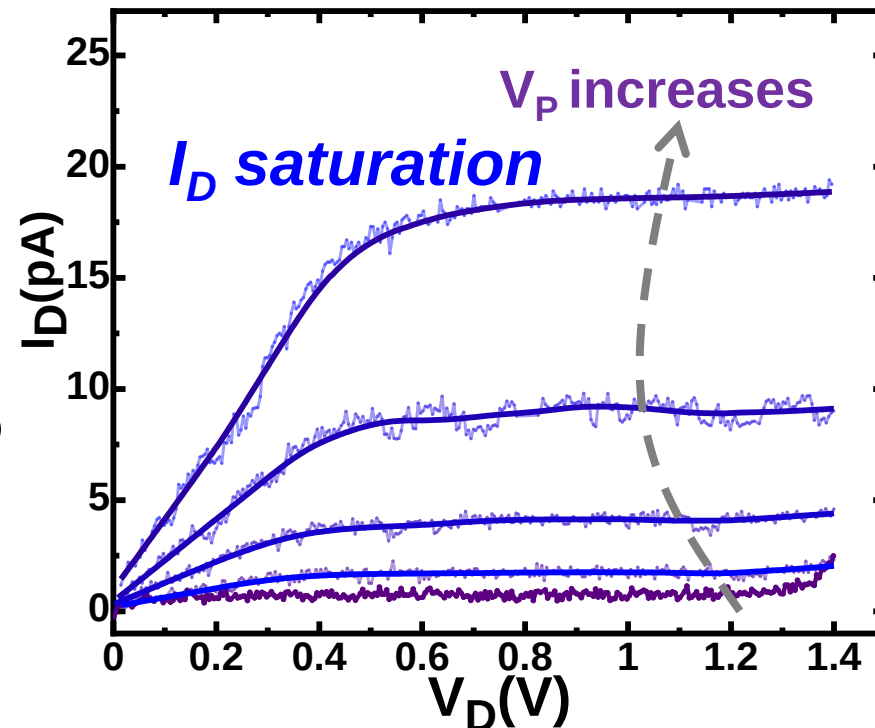
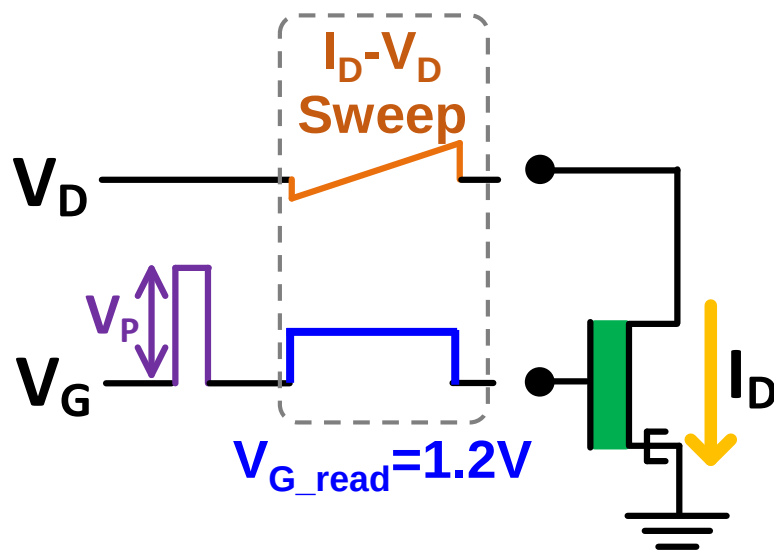
Characteristics of Fe-FinTFET (2/4)

- Program/erase mode with large gate voltage of $|V_p|$
 - Non-volatile ferroelectric (FE) polarization behavior for storage
- Readout mode with small gate-voltage bias (V_{G_read})
 - For programmed state, the read-out $I_D@V_{G_read}=1.2V$ is **lower** than $I_D@V_{G_read}=0V$
 - For erased state, the read-out $I_D@V_{G_read}=1.2V$ is **higher** than $I_D@V_{G_read}=0V$



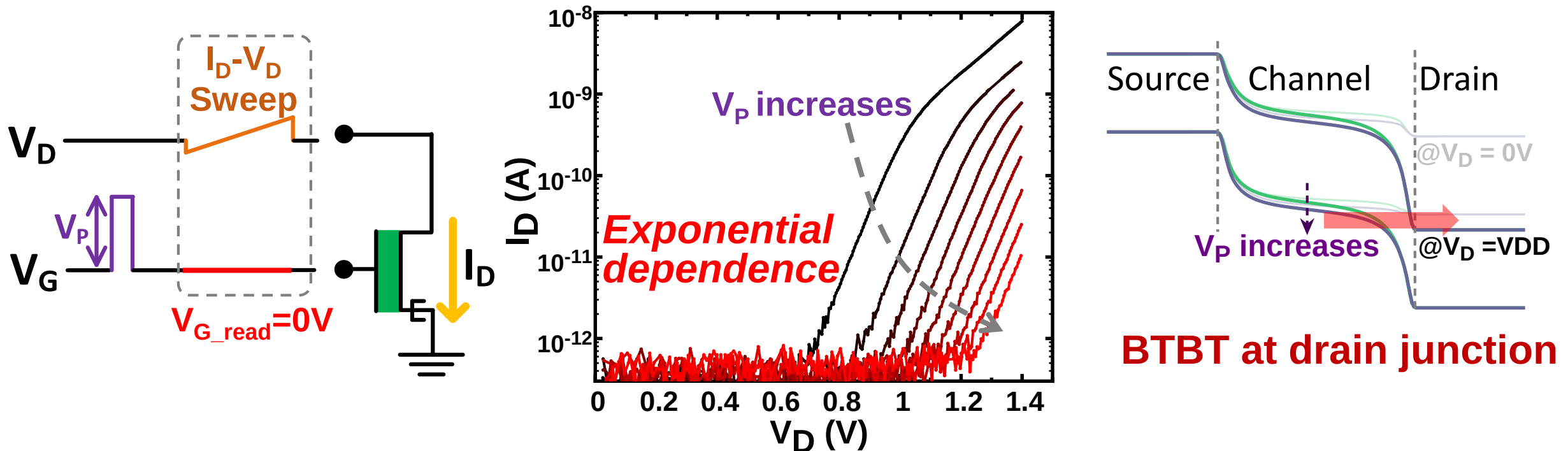
Characteristics of Fe-FinTFET (3/4)

- The BTBT region of source/drain is dependent on V_{G_read} with various V_p
 - For $V_{G_read}=1.2V$: as V_p increases, I_D - V_D curves show saturation at larger V_D like nTFET
 - Indicating that the BTBT behavior occurs at **source junction**



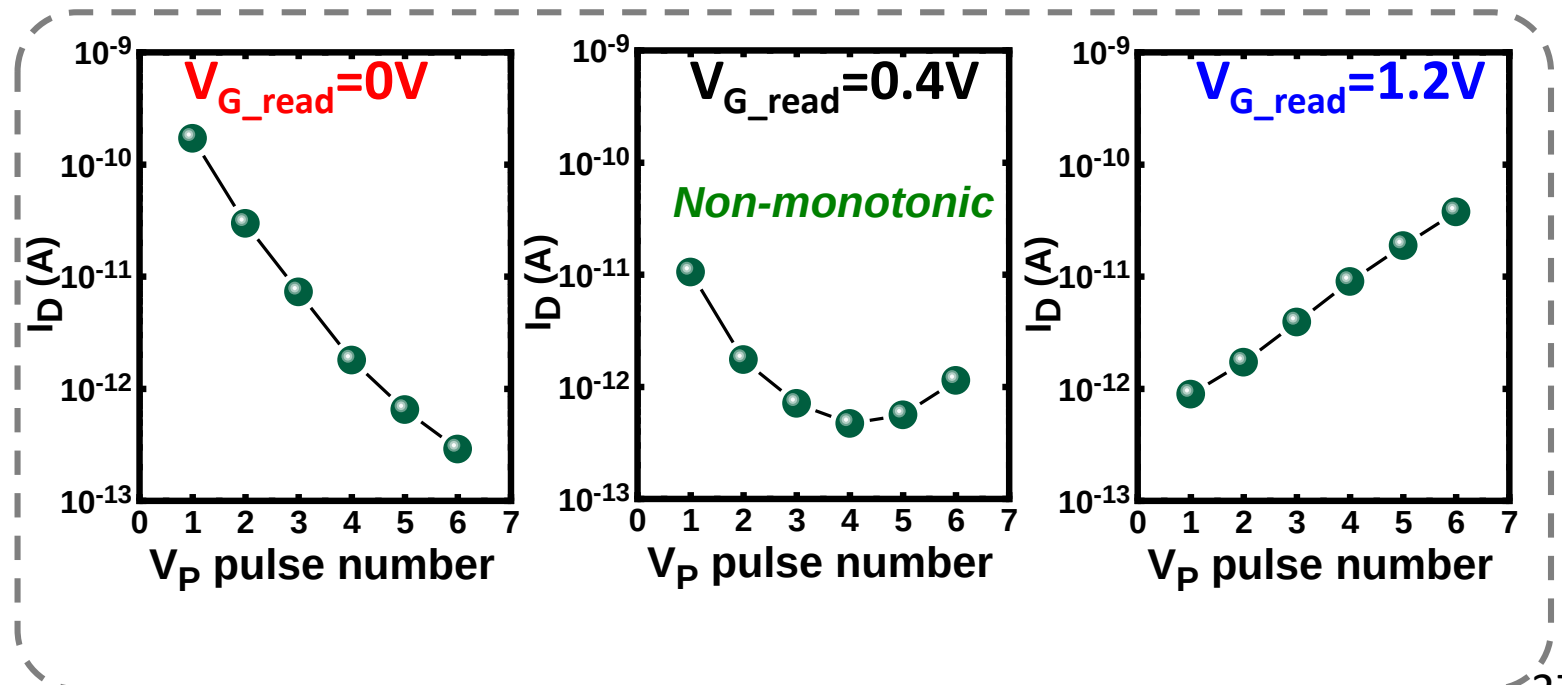
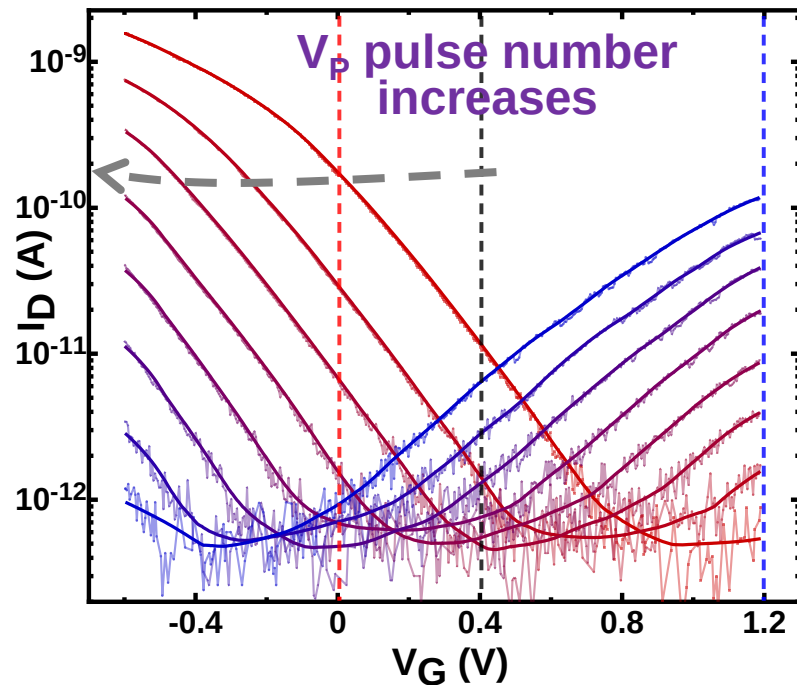
Characteristics of Fe-FinTFET (3/4)

- The BTBT region of source/drain is dependent on V_{G_read} with various V_p
 - For $V_{G_read}=0V$: I_D - V_D curves show exponential dependence
 - Indicating that the BTBT behavior occurs at **drain junction**



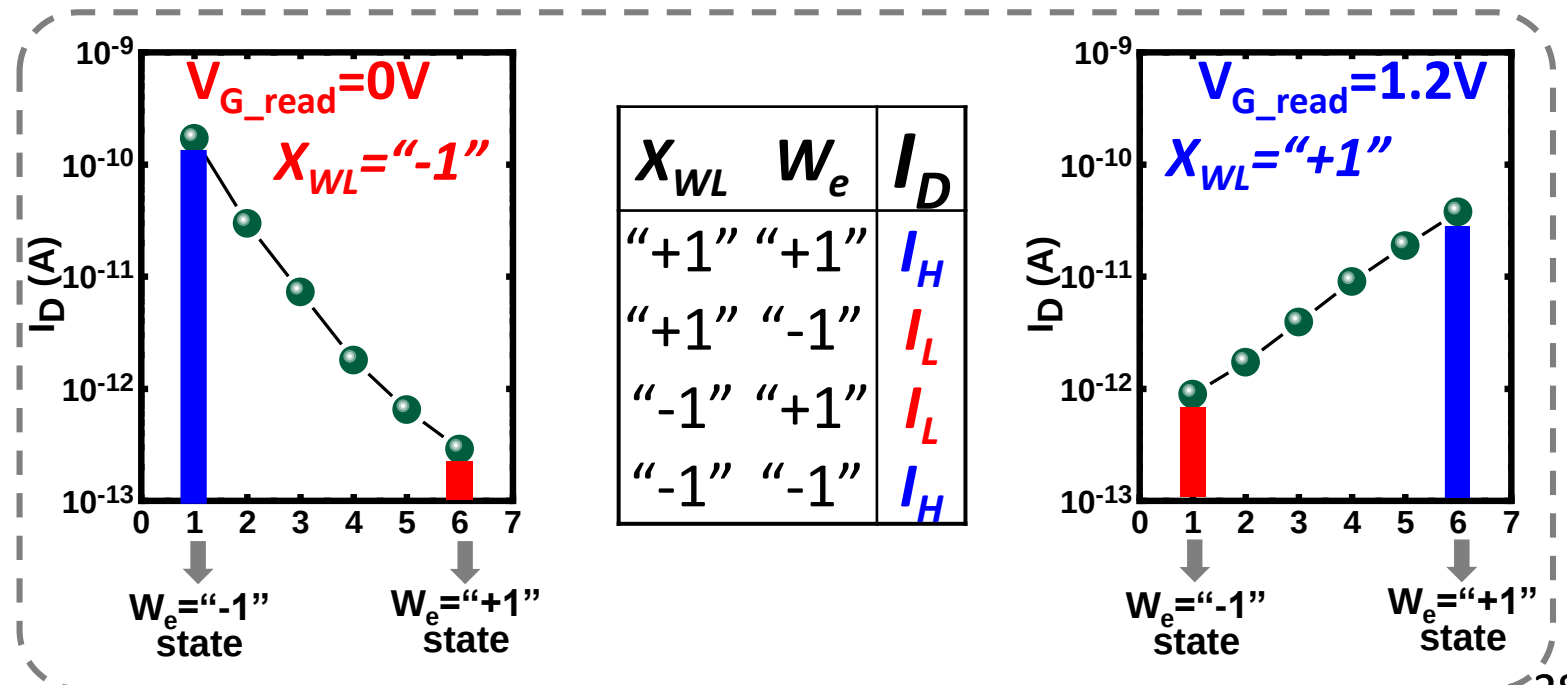
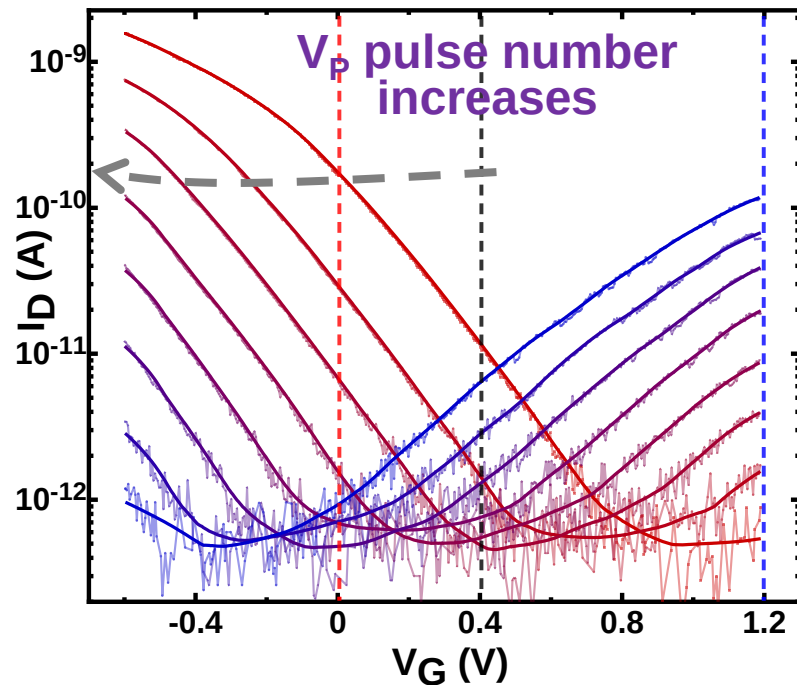
Characteristics of Fe-FinTFET (4/4)

- Ambipolar I_D characteristic for **XNOR operation** in **one Fe-FinTFET**
 - With continuous V_p applied $\rightarrow$ Ambipolar I_D - V_G curve gradually shifts
 - $\rightarrow$ Non-monotonic modulation of I_D , and opposite trend with V_{G_read} of 0V and 1.2V.



Characteristics of Fe-FinTFET (4/4)

- Ambipolar I_D characteristic for **XNOR operation** in **one Fe-FinTFET**
 - With continuous V_p applied $\rightarrow$ Ambipolar I_D - V_G curve gradually shifts
 $\rightarrow$ Non-monotonic modulation of I_D , and opposite trend with V_{G_read} of 0V and 1.2V.
 - Implement **XNOR operation** between X_{WL} and W_e in one single Fe-FinTFET device

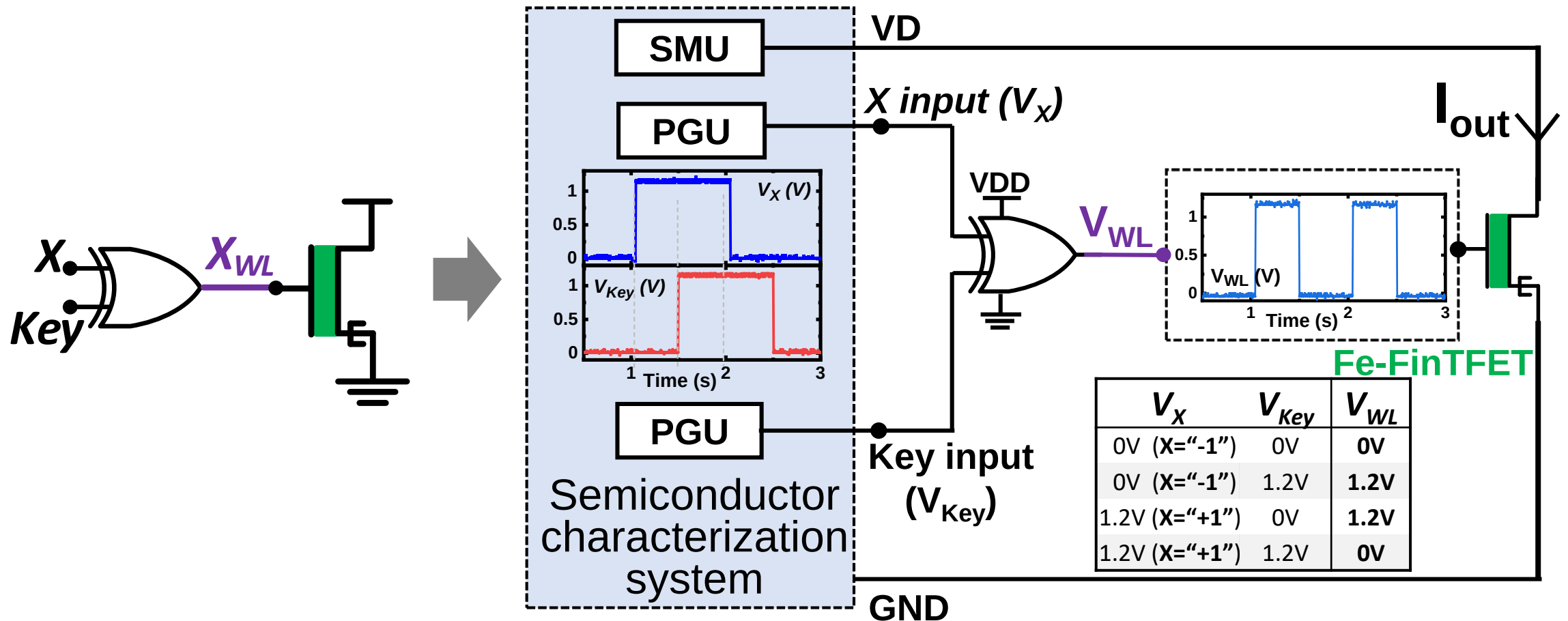


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Measurement setup

- The encrypted multiplication unit based on Fe-FinTFET

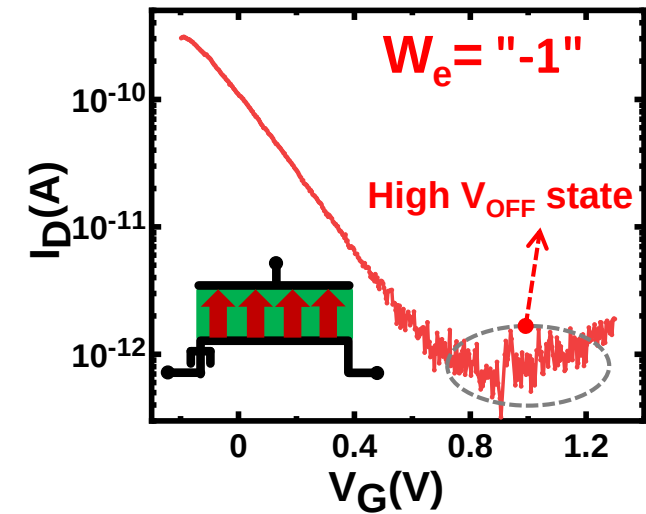
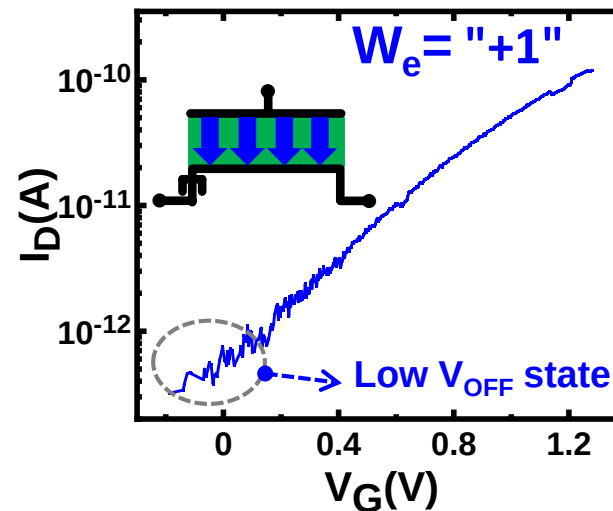


Fe-FinTFET-based encrypted multiplication (1/5)

—Weight storage

- Weight storage stage with encryption
 - Raw W is modulated by XOR-cipher with authorized Key , resulting in W_e
 - W_e of (“+1”, “-1”) is programed into Fe-FinTFET with different V_p

W	Key	W_e	V_{OFF} state
"-1"	0	"-1"	High
" +1"	0	" +1"	Low
"-1"	1	" +1"	Low
" +1"	1	"-1"	High

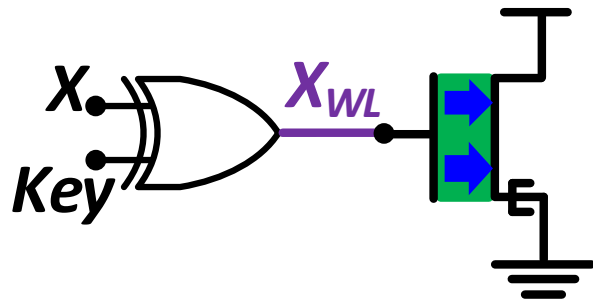


Readout I_D - V_G of Fe-FinTFET with different W_e state

Fe-FinTFET-based encrypted multiplication (2/5)

— Multiplication with implicit decryption

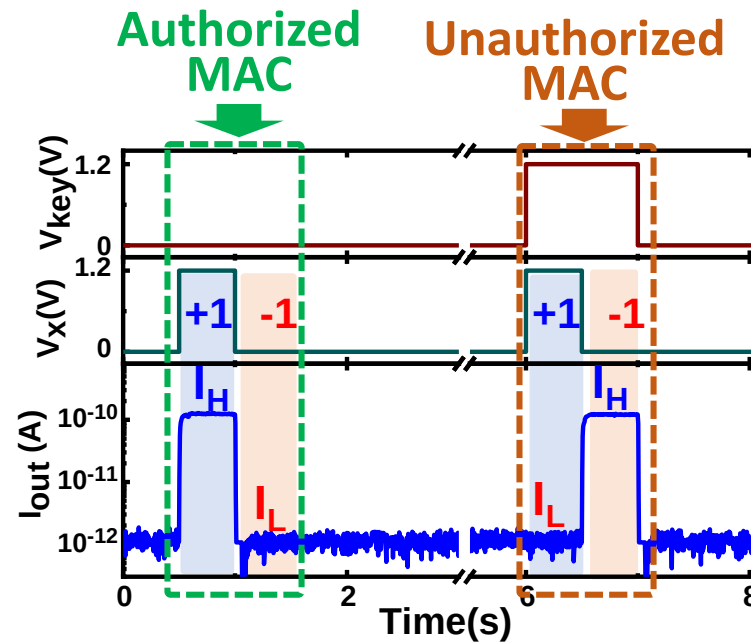
- For raw $W=“+1”$: when authorized $Key = 0 \rightarrow W_e = “+1”$
 - Applying **authorized key=0**: only $X=“+1”$ can make W_e cell fully conductive with I_H
 - Applying **unauthorized wrong key=1**: the multiplication results are wrong



$W=“+1”$, $Key=0$

$W_e=“+1”$

Fe-FinTFET: Low V_{OFF} state

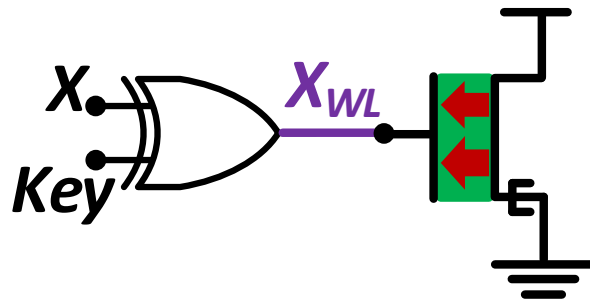


Predefined			IN	Authorized MAC	Unauthorized MAC		
W	Key	W_e	X	Key	I_{out}	Key'	$I_{out\ u}$
" +1 "	0	" +1 "	" -1 "	0	I_L	1	I_H
			" +1 "		I_H		I_L

Fe-FinTFET-based encrypted multiplication (3/5)

— Multiplication with implicit decryption

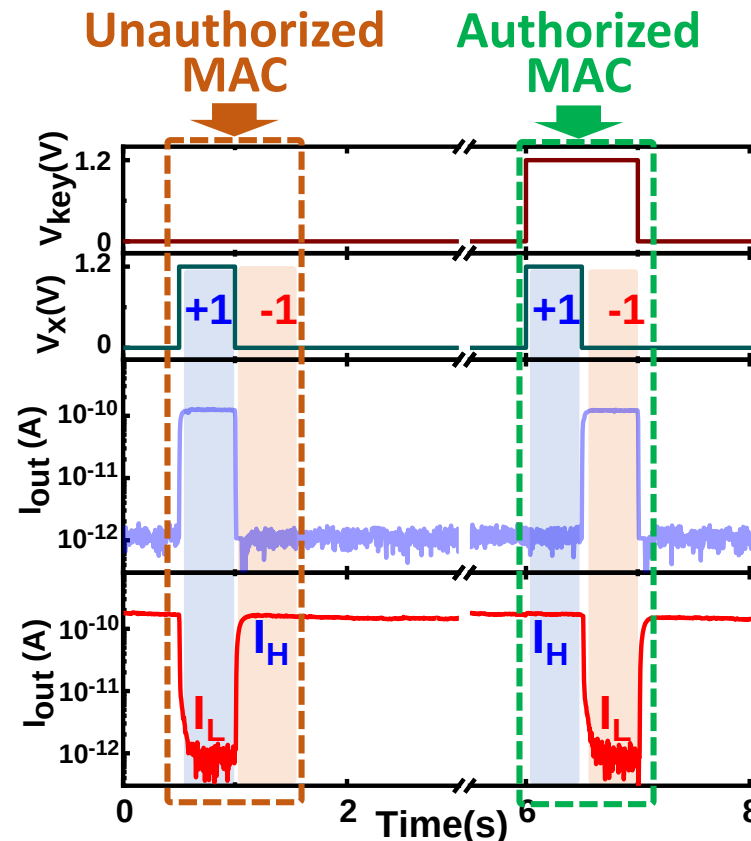
- For raw $W=“+1”$: when authorized $Key = 1 \rightarrow W_e = “-1”$
 - Applying **authorized key=1**: only $X=“+1”$ makes the W_e cell fully conductive with I_H
 - Applying **unauthorized wrong key=0**: the multiplication results are wrong



$W=“+1”$, $Key=1$

$W_e=“-1”$

Fe-FinTFET: **High V_{OFF} state**

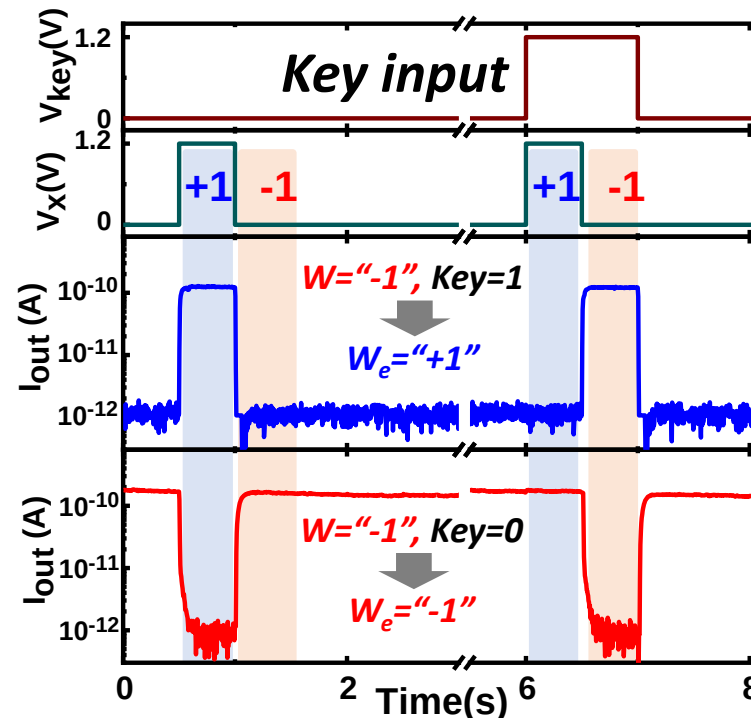
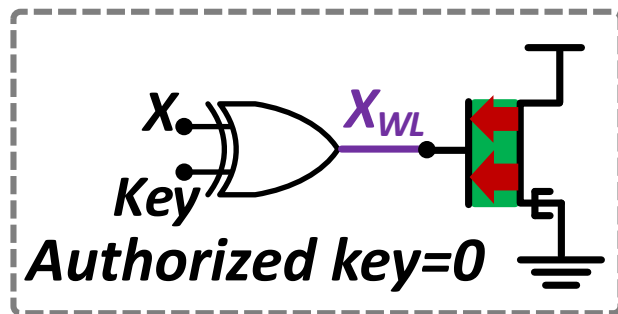
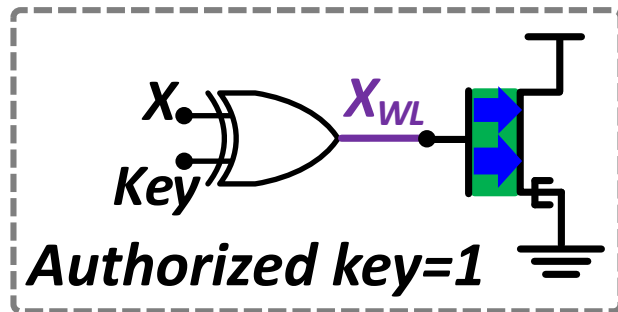


Predefined		IN		Authorized MAC		Unauthorized MAC	
W	Key	W_e	X	Key	I_{out}	Key'	$I_{out u}$
" +1 "	0	" +1 "	" -1 "	0	I_L	1	I_H
" +1 "	1	" -1 "	" +1 "	1	I_H	0	I_L

Fe-FinTFET-based encrypted multiplication (4/5)

— Multiplication with implicit decryption

- For raw $W=“+1”$:
- For raw $W=“-1”$: The encrypted multiplication process is similar to the case of $W=“+1”$
- When and only when the *key* is right, the multiplication is carried out properly

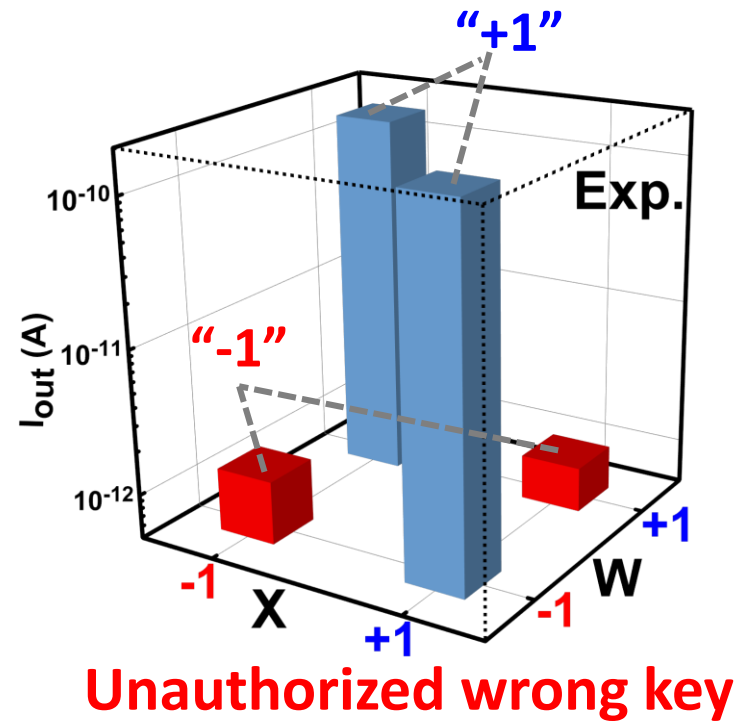
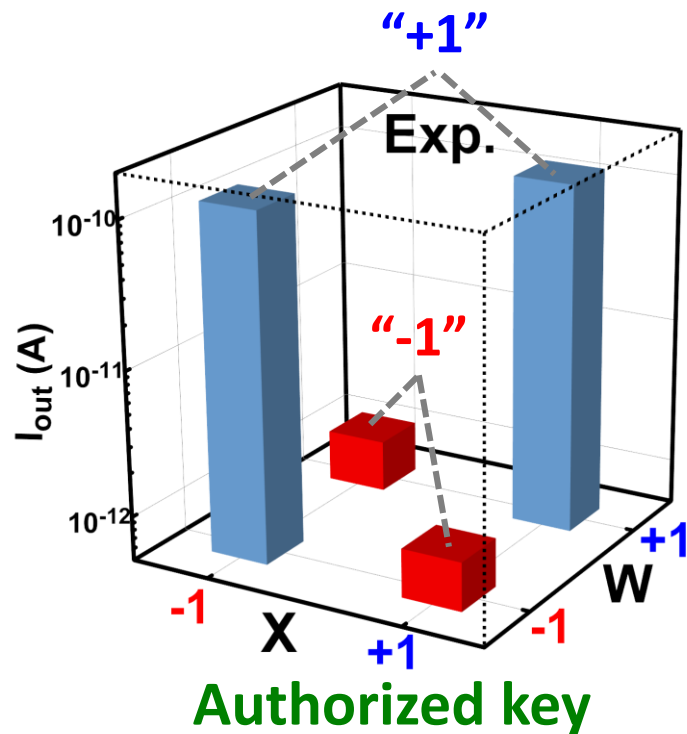


Predefined			IN	Authorized MAC		Unauthorized MAC	
W	Key	W_e	X	Key	I_{out}	Key'	$I_{out\ u}$
" +1 "	0	" +1 "	" -1 " " +1 "	0	I_L I_H	1	I_H I_L
" +1 "	1	" -1 "	" -1 " " +1 "	1	I_L I_H	0	I_H I_L
" -1 "	0	" -1 "	" -1 " " +1 "	0	I_H I_L	1	I_L I_H
" -1 "	1	" +1 "	" -1 " " +1 "	1	I_H I_L	0	I_L I_H

Fe-FinTFET-based encrypted multiplication (5/5)

— Multiplication with implicit decryption

- Implement multiplication with implicit decryption in **one Fe-FinTFET W_e cell**

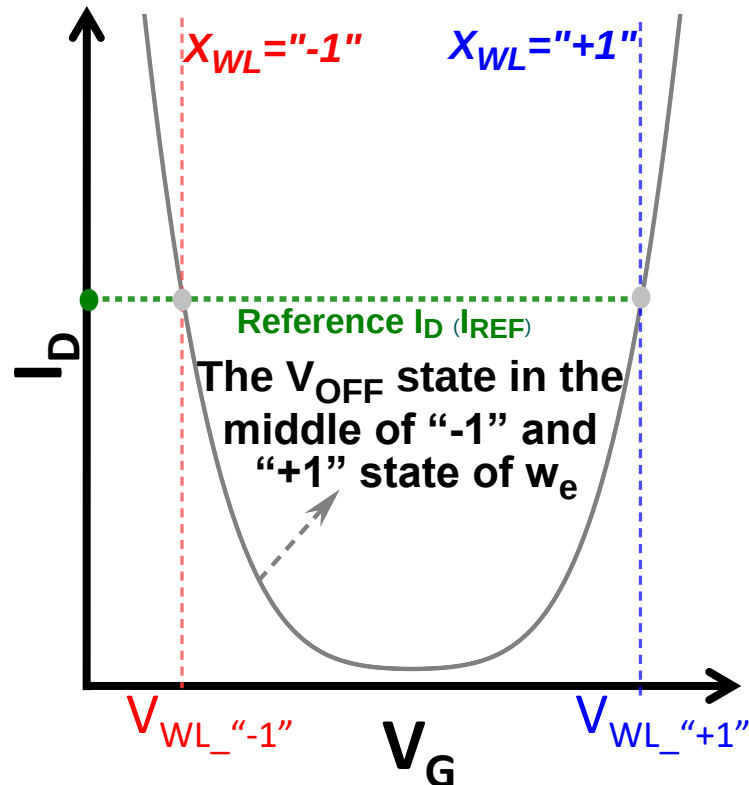


Predefined	IN	Authorized MAC	Unauthorized MAC
W Key W_e	X	Key I_{out}	Key' $I_{out u}$
"+1" 0 "+1"	"-1"	0 I_L	1 I_H
"+1" 1 "-1"	"-1"	1 I_L	0 I_H
"-1" 0 "-1"	"-1"	0 I_H	1 I_L
"-1" 1 "+1"	"-1"	1 I_H	0 I_L

Only with authorized key can the results of **XNOR-based signed multiplication** be correct, otherwise, the results will be wrong and have the opposite sign.

Fe-FinTFET-based eMAC with multilevel weight (1/4)

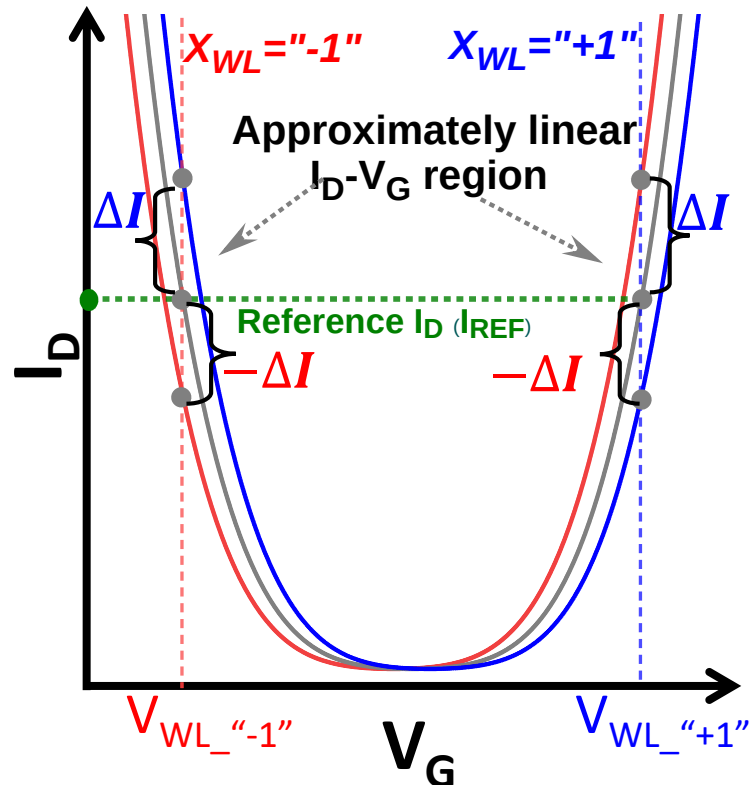
- For $W_e = "0"$:
 - The FE state is modulated to ensure the V_{OFF} of FeTFET is in the middle of $"-1"$ and $" +1"$ state of W_e , and thus the I_D is the same when X_{WL} is $"-1"$ or $" +1"$.
 - I_D at this case is the reference current I_{REF} , representing the multiplication result of $"0"$.



$$I_D (@X_{WL} = "-1") = I_D (@X_{WL} = "+1") = I_{REF}$$

Fe-FinTFET-based eMAC with multilevel weight (2/4)

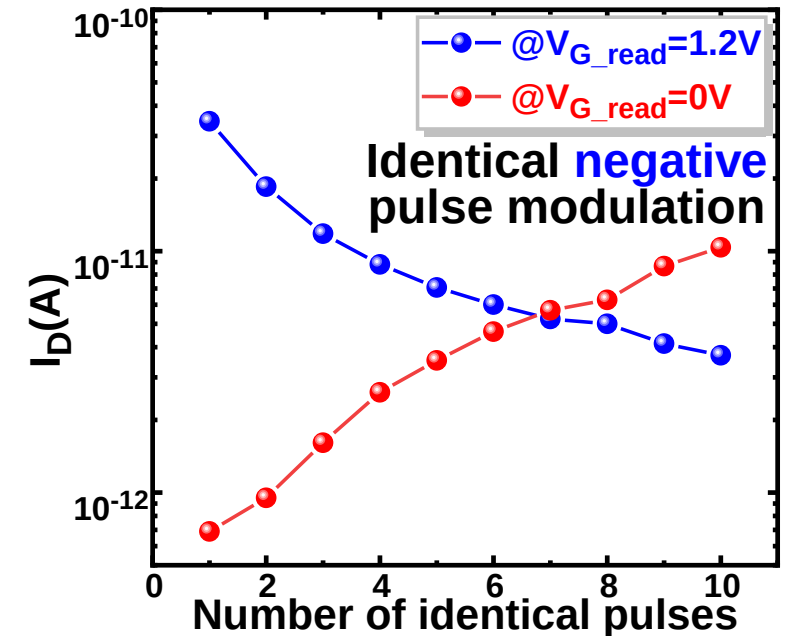
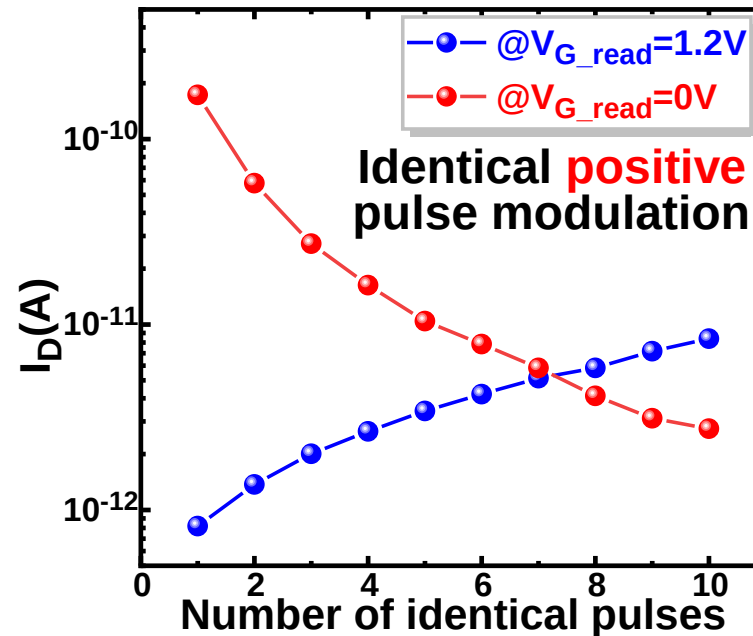
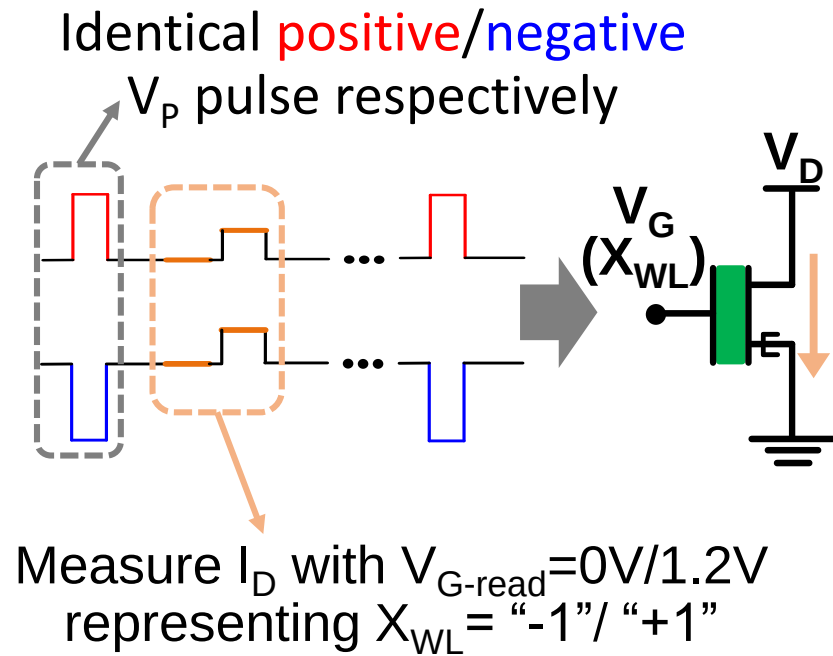
- Continuously modulated W_e with multilevel state
 - Shift I_D - V_G curve with partial FE polarization switching → The change of I_D (ΔI) for X_{WL} of “-1”/“+1” is symmetrical within the approximately linear region
 - The analog ΔI represents multiplication results of X_{WL} and W_e with multilevel state



X_{WL}	ΔI representing $X_{WL} \cdot W_e$	
	For positive W_e (leftward shift I_D - V_G)	For negative W_e (rightward shift I_D - V_G)
"-1"	$-\Delta I$	ΔI
" +1"	ΔI	$-\Delta I$

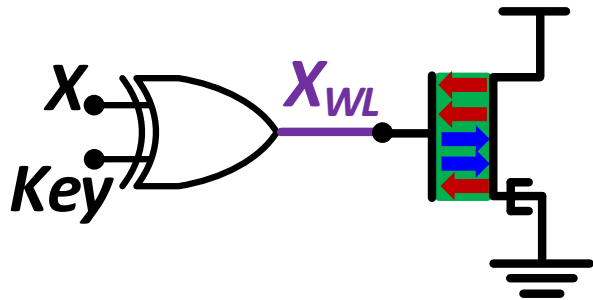
Fe-FinTFET-based eMAC with multilevel weight (3/4)

- Gradually modulated drain current for multilevel W_e storage
 - With identical pulses applied to the gate of Fe-FinTFET, the FE polarization can be gradually switched in succession, resulting in the continuous modulation of ΔI

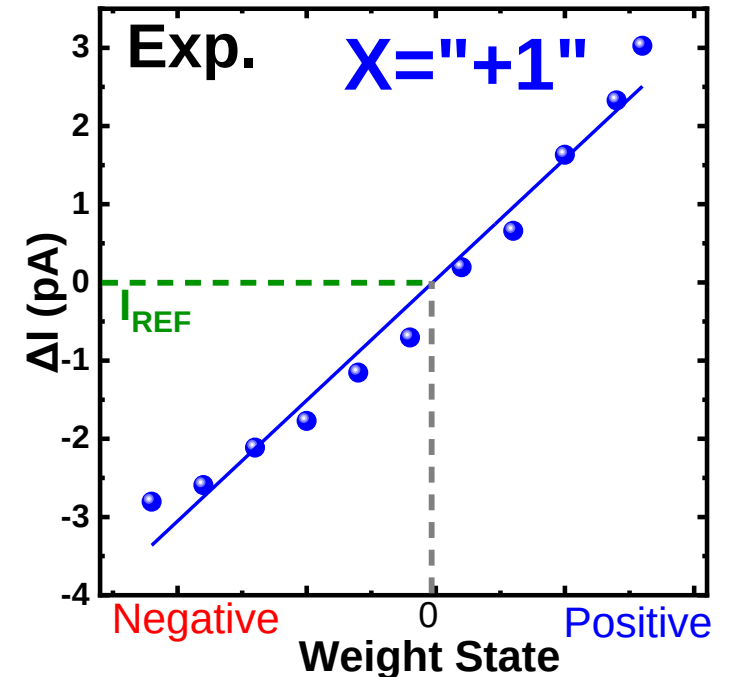
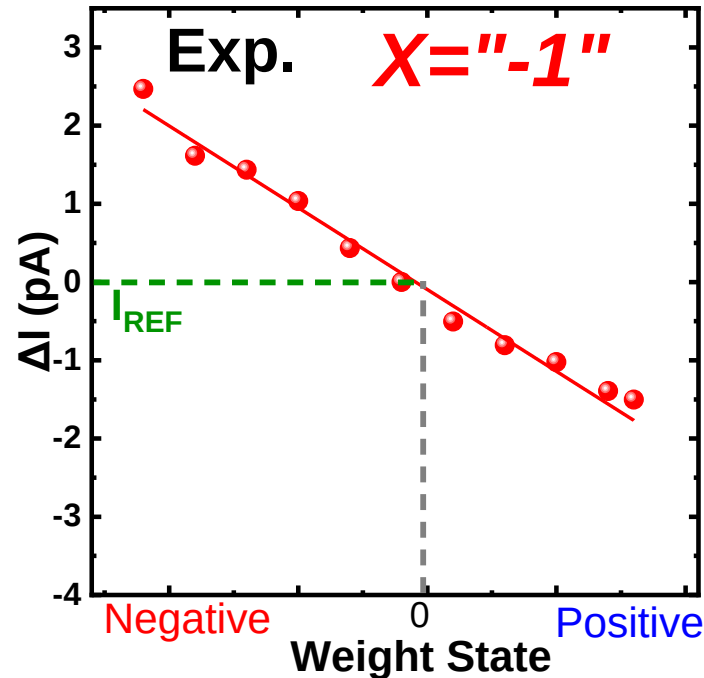


Fe-FinTFET-based eMAC with multilevel weight (4/4)

- Experimentally demonstrate encrypted multiplication of **signed X** and **multilevel W** in **one Fe-FinTFET W_e cell** with further improved density

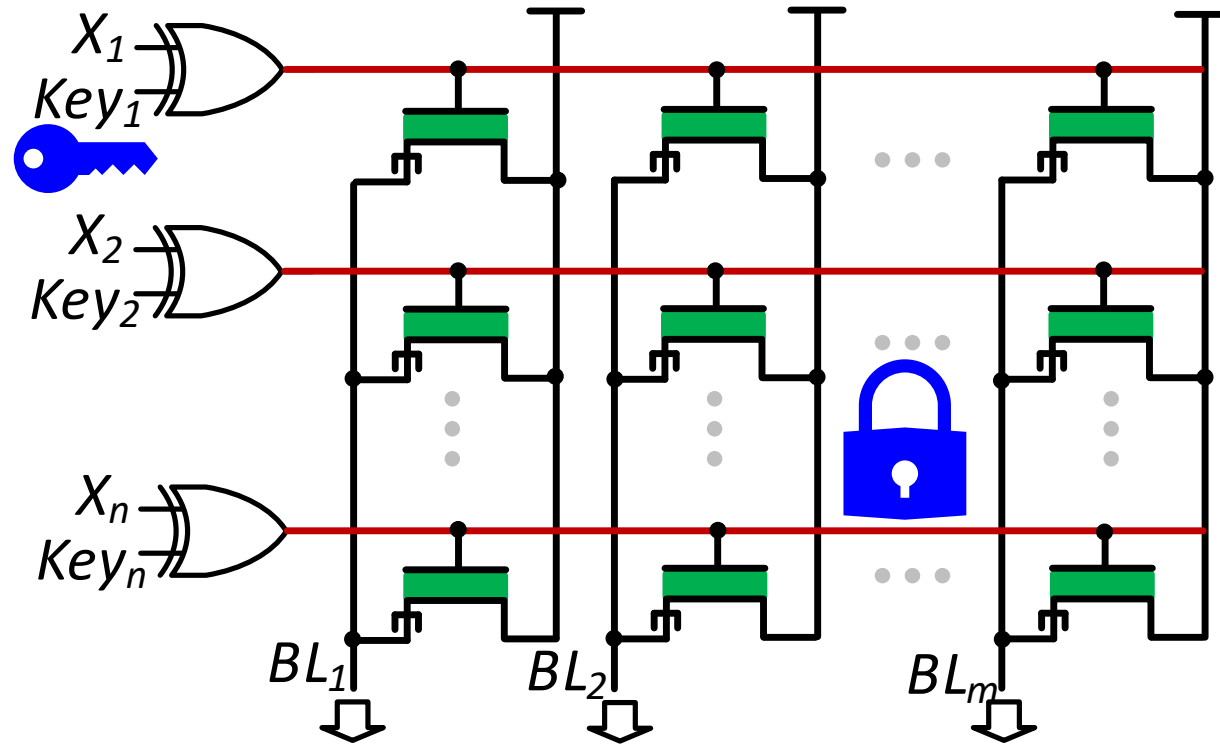


- ✓ Signed X of “-1”/ “+1”
- ✓ Digital Key bit
- ✓ Multilevel weight (W)



Multiplication results with authorized key and signed input

FeTFET array for highly-parallel encrypted MAC (eMAC)



Partial weighted sum (y_j) as I_{BL_j}

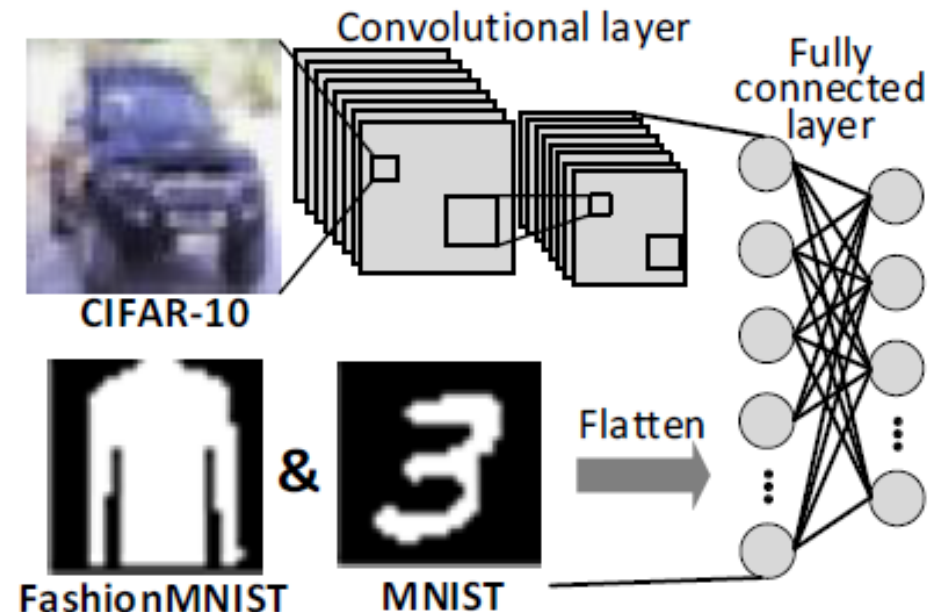
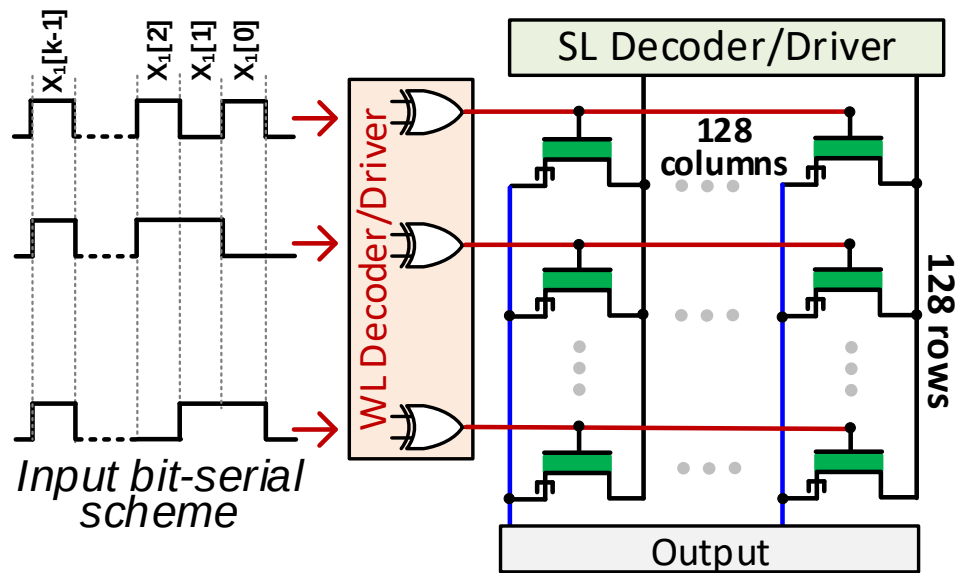
$$y_j = \sum_i x_i \times w_{ij}$$

Outline

- Design of FeTFET-based encrypted MAC (eMAC)
- Experiments of FeTFET-based eMAC behavior
- FeTFET-based encrypted nvCIM macro for edge secure AI
 - Encrypted inference and learning process of ANNs
 - eMAC array performance evaluation
- Summary

FeTFET-based encrypted nvCIM

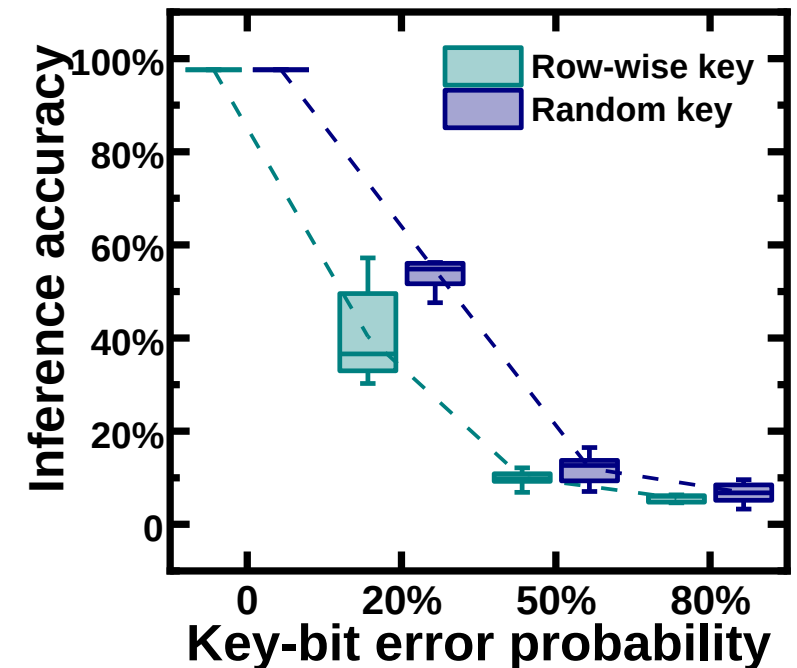
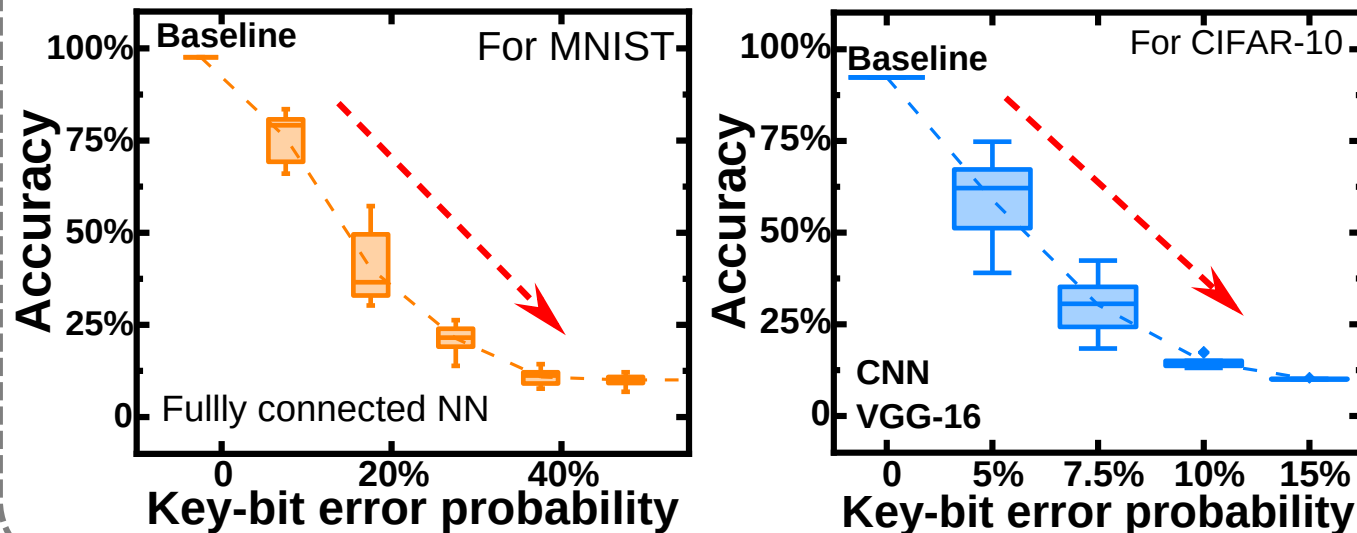
- ANN structure based on proposed FeTFET-based encrypted nvCIM macro
 - eMAC based on FeTFET array of multilevel weight, and inputs with bit-serial extension
 - ANN models with fully connected layers and convolutional layers can be encrypted



FeTFET-based encrypted nvCIM —Encrypted inference

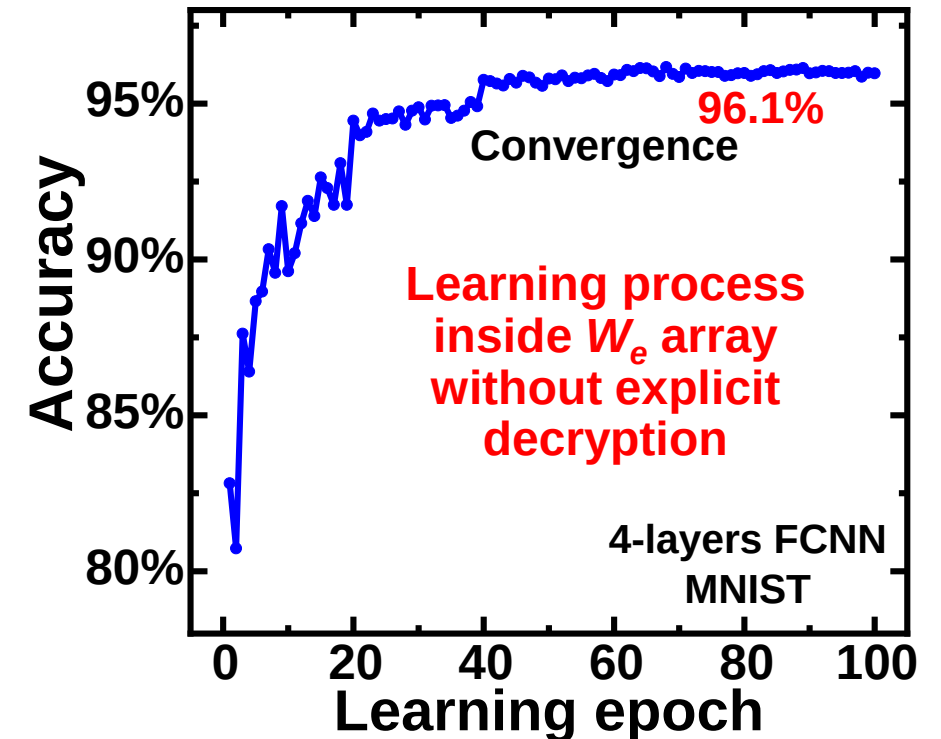
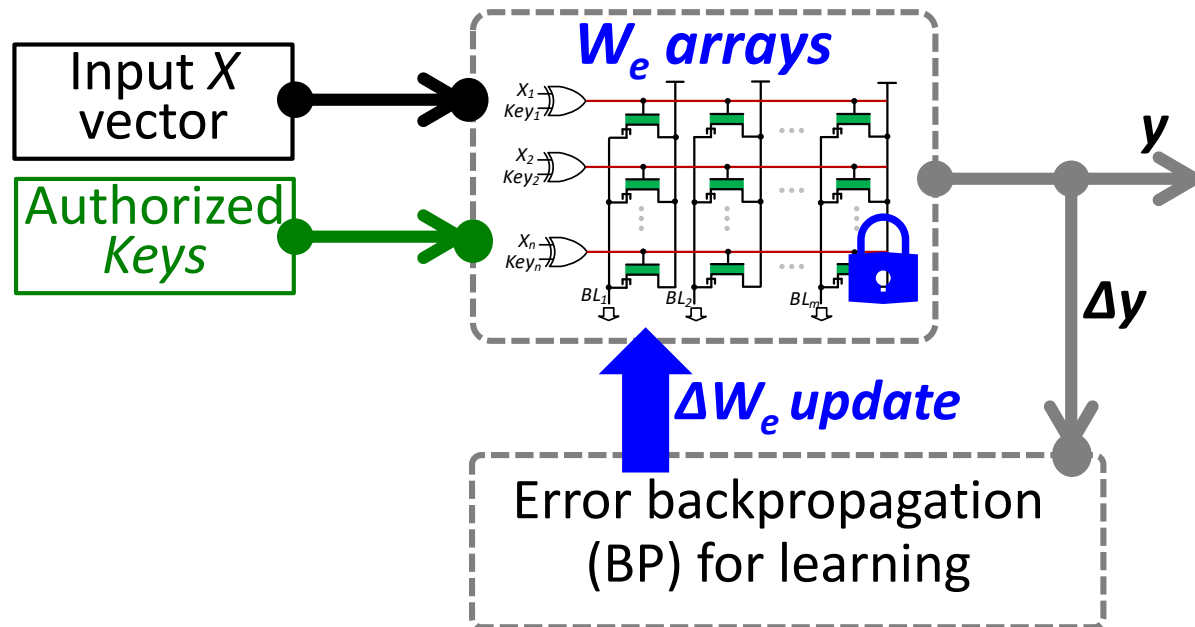
- Encrypted inference process with authorized-key for private ANNs
 - Prevent unauthorized utilization of ANN model
 - Row-wise weight encryption enables considerable encryption effectiveness compared with conventional random encryption.

Severely degradation of inference accuracy when using random guess on the keys



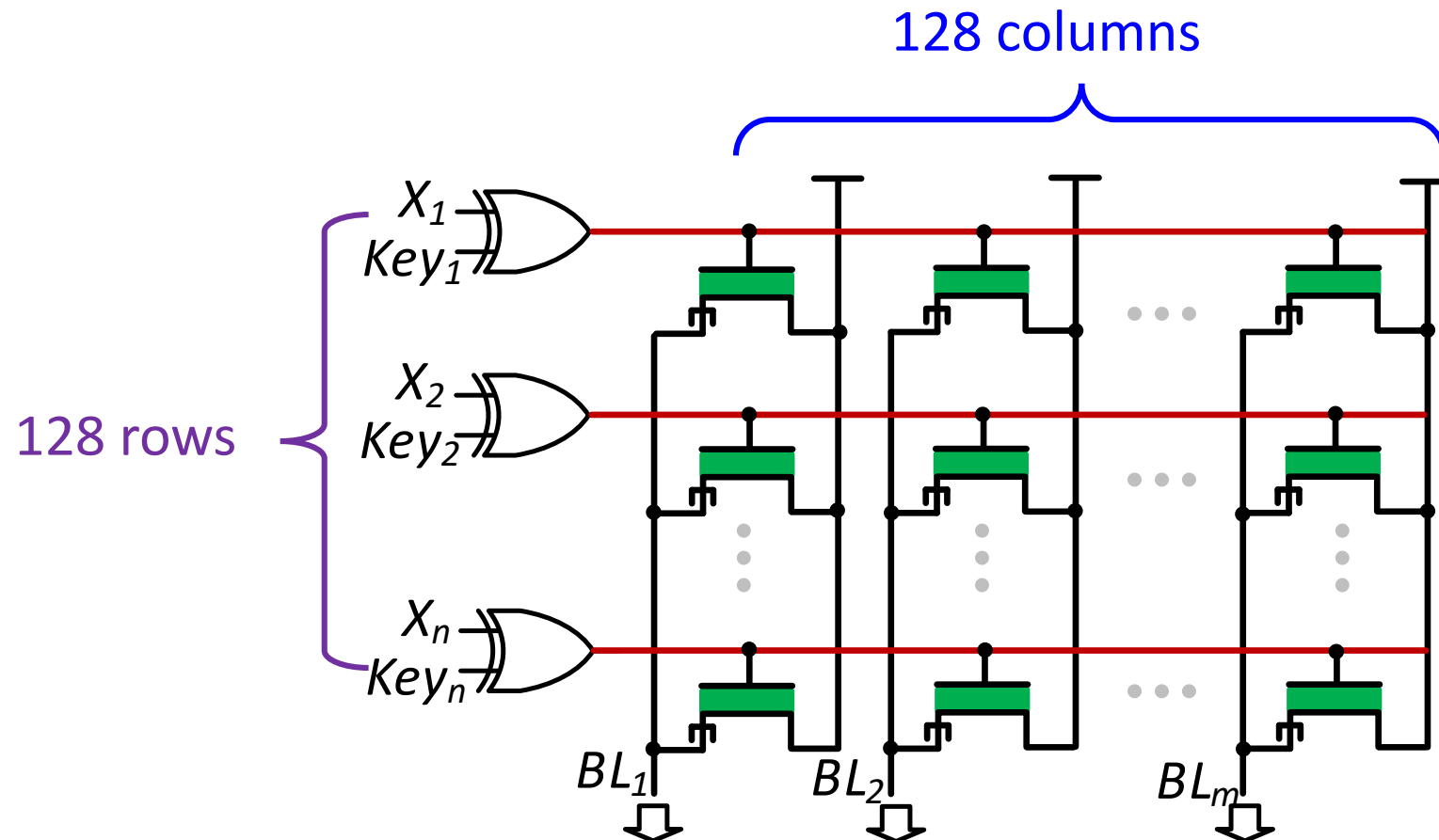
FeTFET-based encrypted nvCIM —Encrypted learning

- Encrypted learning process with authorized keys
 - When using authorized key, the *in-situ* learning process can be implemented **directly inside encrypted weight (W_e) array** without explicit decryption



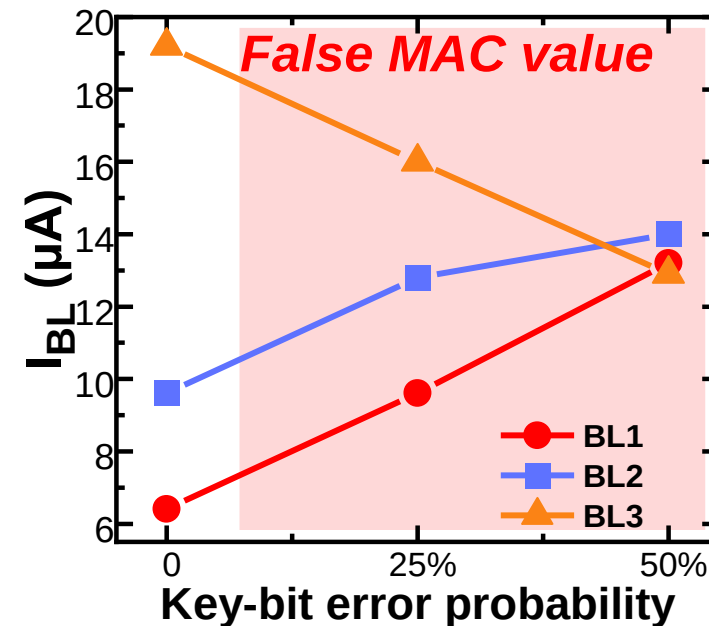
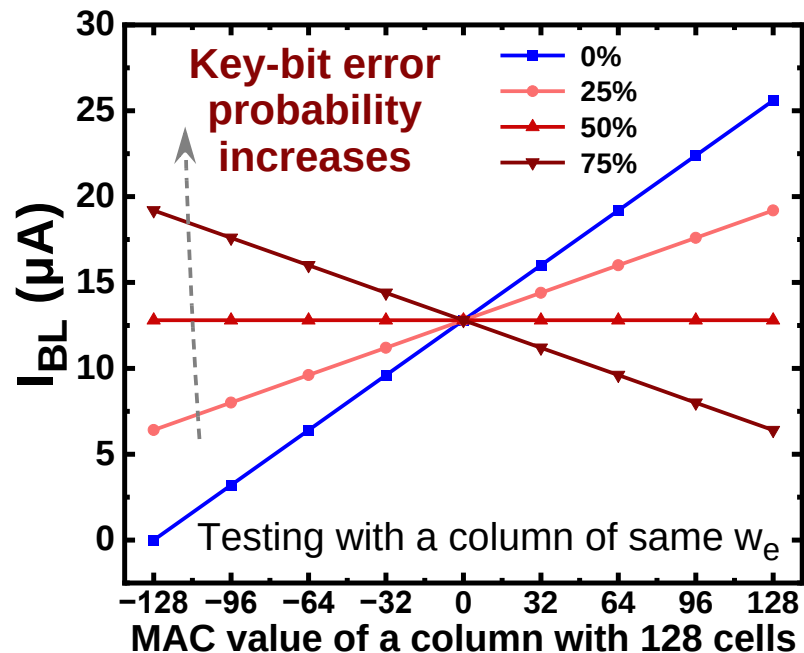
eMAC array performance evaluation

- FeTFET-based eMAC array based on developed FeTFET model



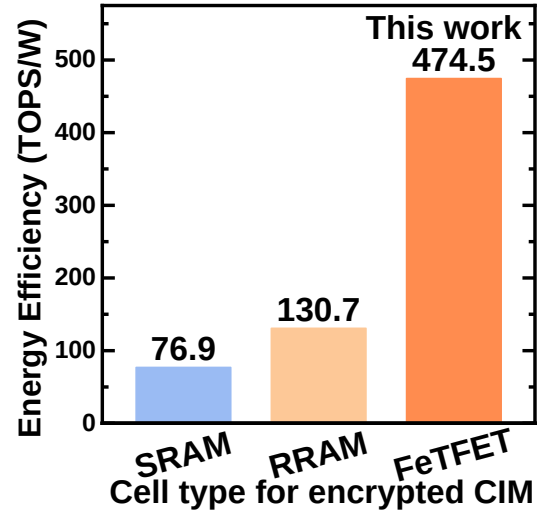
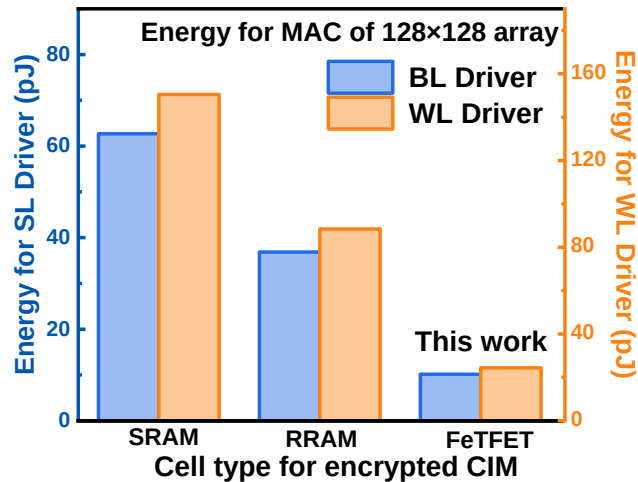
eMAC array performance evaluation

- FeTFET array based encrypted MAC computing
 - With fully authorized key, I_{BL} shows good linearity with mathematic MAC results
 - With key-bit error probability increasing, the deviation of I_{BL} increases, which results in the wrong MAC, enabling private ANNs for secure AI.



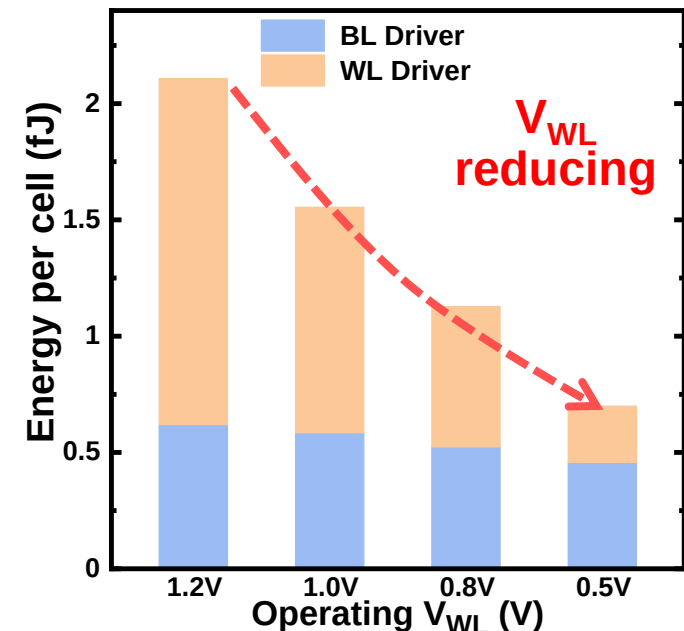
eMAC array performance evaluation

- Energy efficiency of proposed FeTFET-based eMAC
 - Significant improvement in energy efficiency over previous encrypted CIM^{[1]-[2]} due to co-design of encryption algorithm and nvCIM hardware with 1FeTFET W_e cell
 - The energy efficiency can be further improved by reducing V_{WL} while maintaining large on/off ratio due to the advantage of steep subthreshold swing (SS) in TFET



✓ This work: eliminating complementary form of W_e storage, word-line and bit-line

[1] S. Huang, et al., ICCAD 2020. [2] W. Li, et al., CICC, 2021.



Benchmark of proposed FeTFET-based encrypted nvCIM with other reported solutions

	ISSCC 2022 [1]	ICCAD 2020 [2]	CICC 2021 [3]	This work
Cell type	MRAM	SRAM	RRAM	FeTFET
Encrypted weight cell size	1T+1MTJ	6T (Split WL)	2T+2R	1T
Multi-level weight	No	No	--	Yes
MAC type	Digital MAC	Unsigned-AND type		Signed XNOR
Decryption	Decryption circuit	<i>In-situ</i> (Complementary circuit branch)		<i>In-situ</i>
eMAC parallelism	Low	High	Medium	High
ANN application	--	Inference	Inference	Inference & Learning
eMAC energy efficiency (TOPS/W)	127 ^(a)	76.8 ^(b)	130.7 ^(b)	473 TOPS/W ^(b)

(a) Extract performance with 8bIN-4bW in 50% input sparsity [1]. (b) Calculated in Hspice based on the encrypted array of the same size with respectively proposed encrypted weight cell size.

[1] Y. Chiu, et al., ISSCC, pp.178-180, 2022. [2] S. Huang, et al., ICCAD 2020. [3] W. Li, et al., CICC, 2021.

Outline

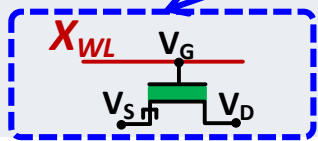
- Design of FeTFET-based encrypted MAC (eMAC)
- Experiments of FeTFET-based eMAC behavior
- FeTFET-based encrypted nvCIM macro for edge secure AI
- **Summary**

Summary

A novel nvCIM with encrypted weight (W_e) of **only one Fe-FinTFET**, enabling ***in-situ* decryption** with high area- and energy-efficiency

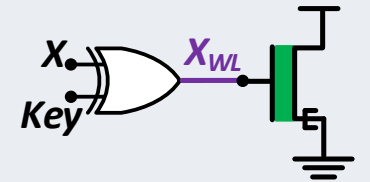
Co-design of eMAC algorithm & FeTFET physics for encrypted nvCIM

- Algorithmically *merge XOR-cipher into XNOR-MAC* by associative law, enabling *in-situ* eMAC in W_e cell of XNOR operation
- Novel FeTFET physics for W_e : *non-volatile polarization* for W_e storage, *gated ambipolar BTBT* for XNOR operation of W_e cell

$$\begin{aligned} y_j &= \sum_i^n x_i \odot w_{i,j} \\ &= \sum_i^n x_i \odot (w_{e,i,j} \oplus \text{Key}_i) \\ &= \sum_i^n (x_i \oplus \text{Key}_i) \odot w_{e,i,j} \end{aligned}$$


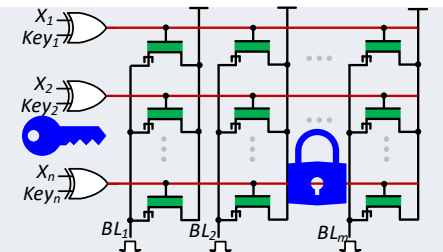
Experimental demonstration of Fe-FinTFET-based eMAC

- W_e cell with XNOR operation for implicit decryption within **one Fe-FinTFET**
- Encrypted multiplication with **multilevel weight** for eMAC with density boost



FeTFET-based encrypted nvCIM for edge secure AI

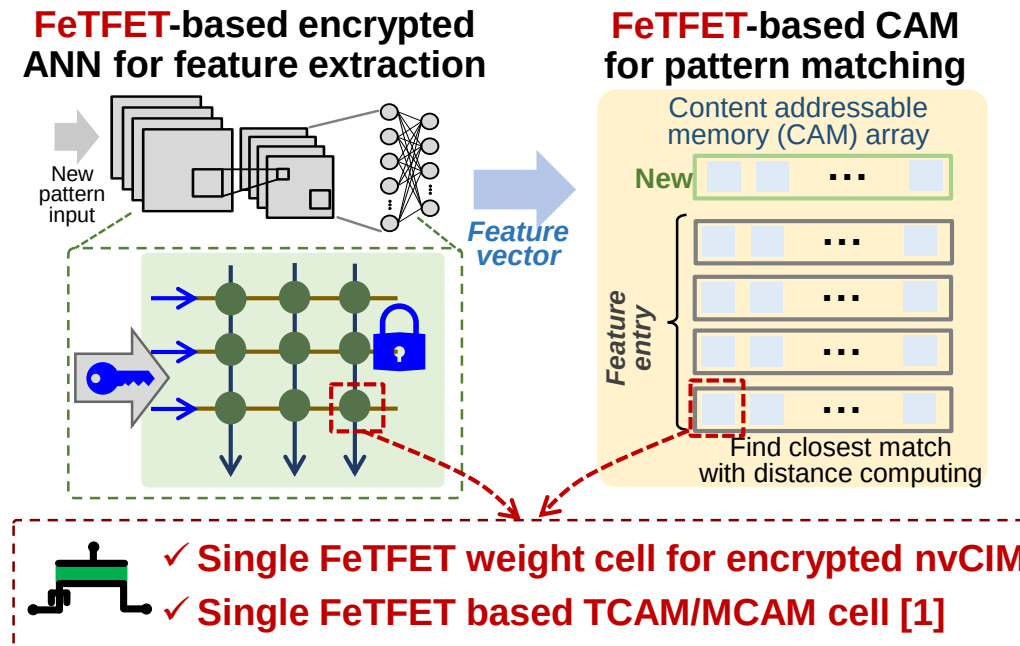
- Encrypted inference and learning process of ANNs
- High area- and energy-efficiency**



Thank you for your attention!

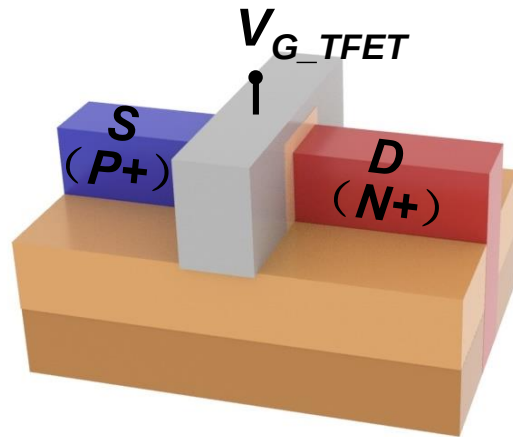
Backup

Memory-augmented NN for one-shot learning

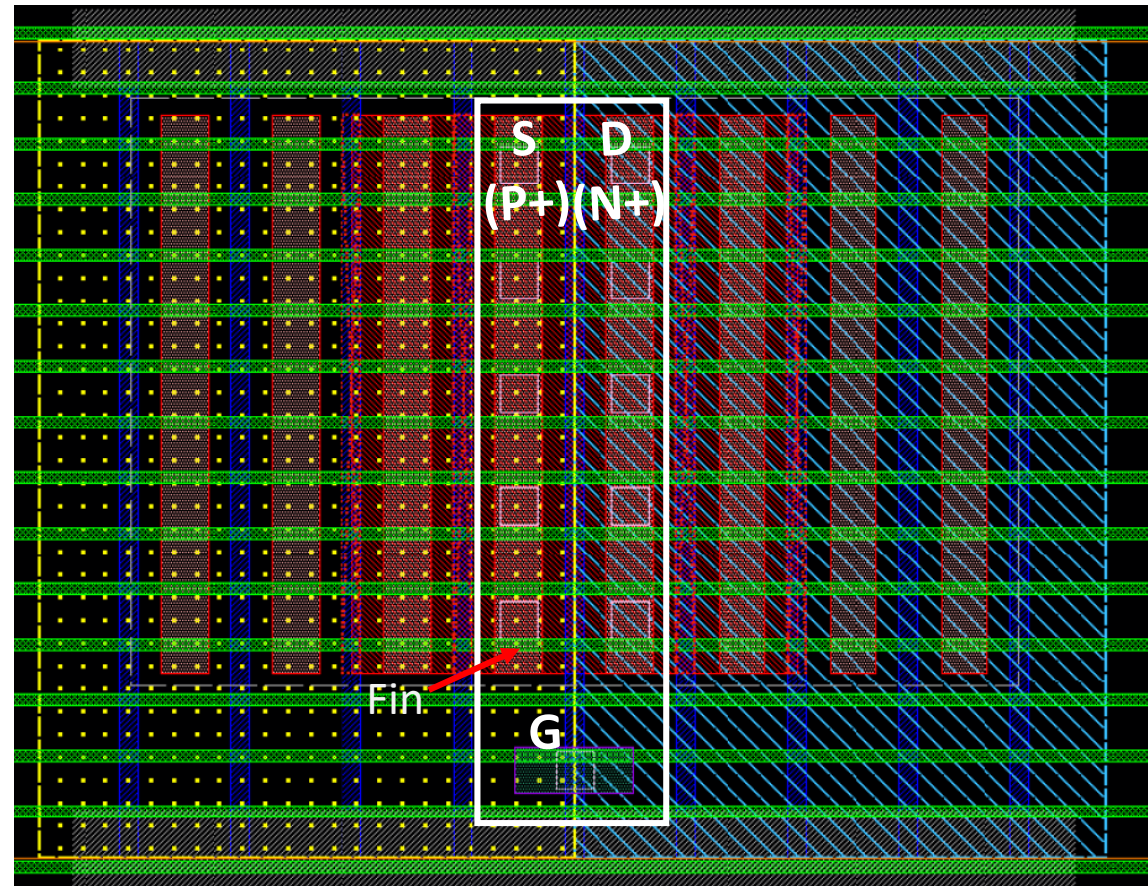


[1] Jin Luo, Weikai Xu et al., IEEE VLSI, 2022.

Fabrication of FinTFET

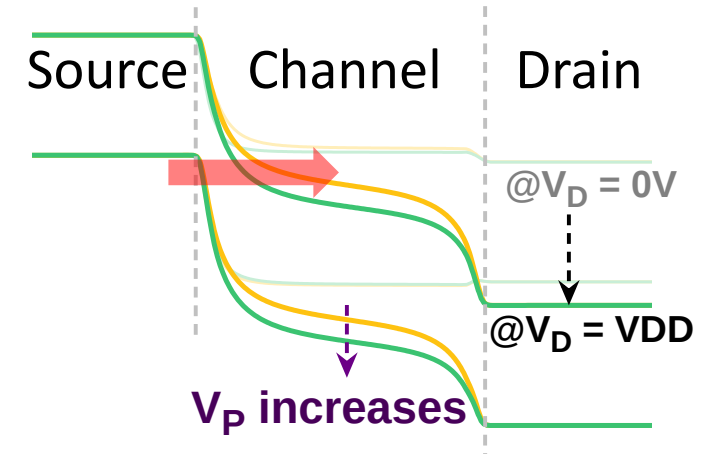
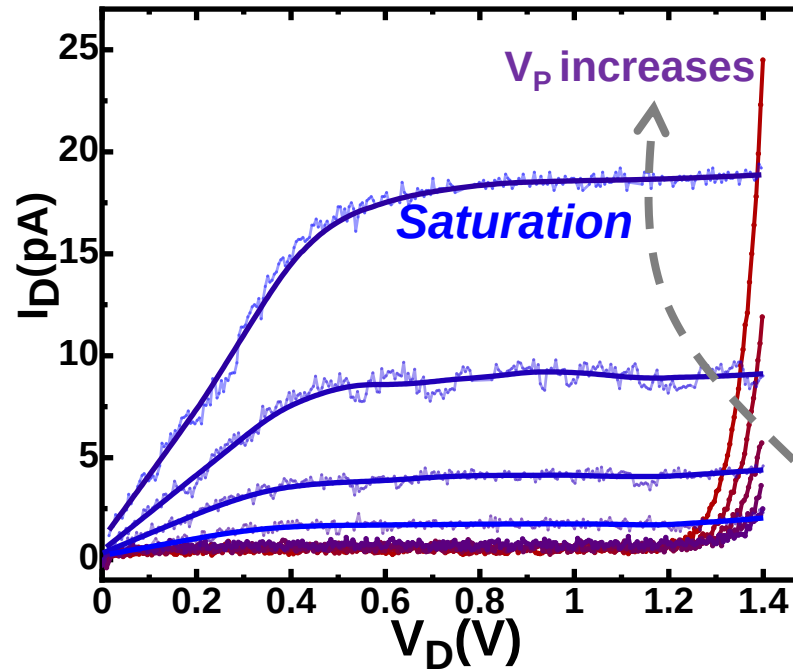
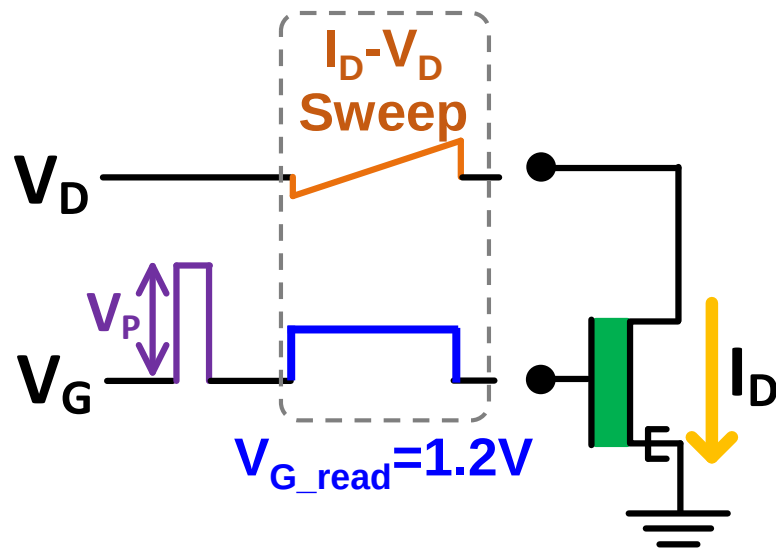


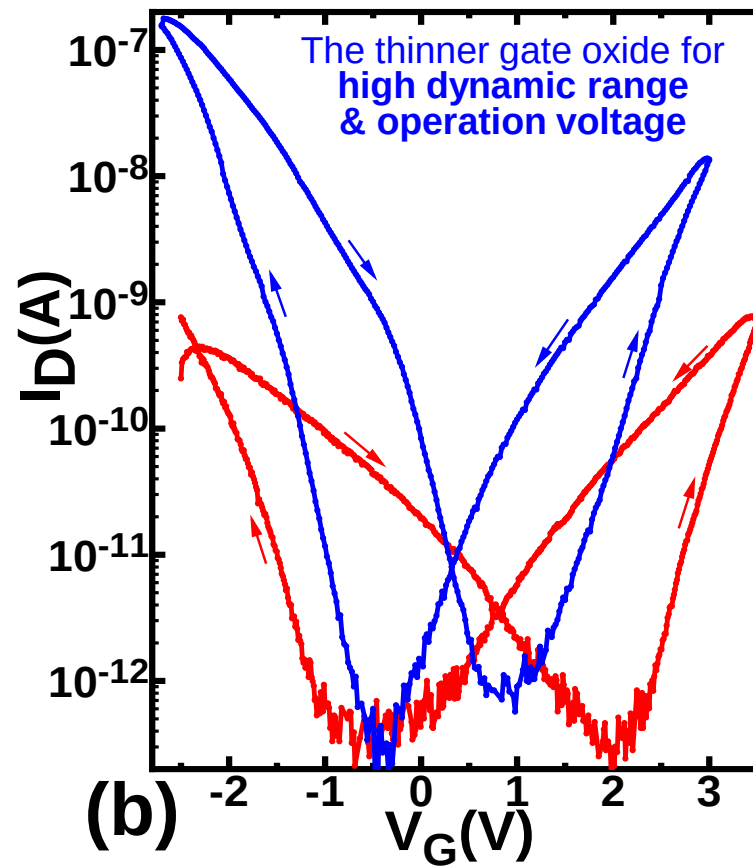
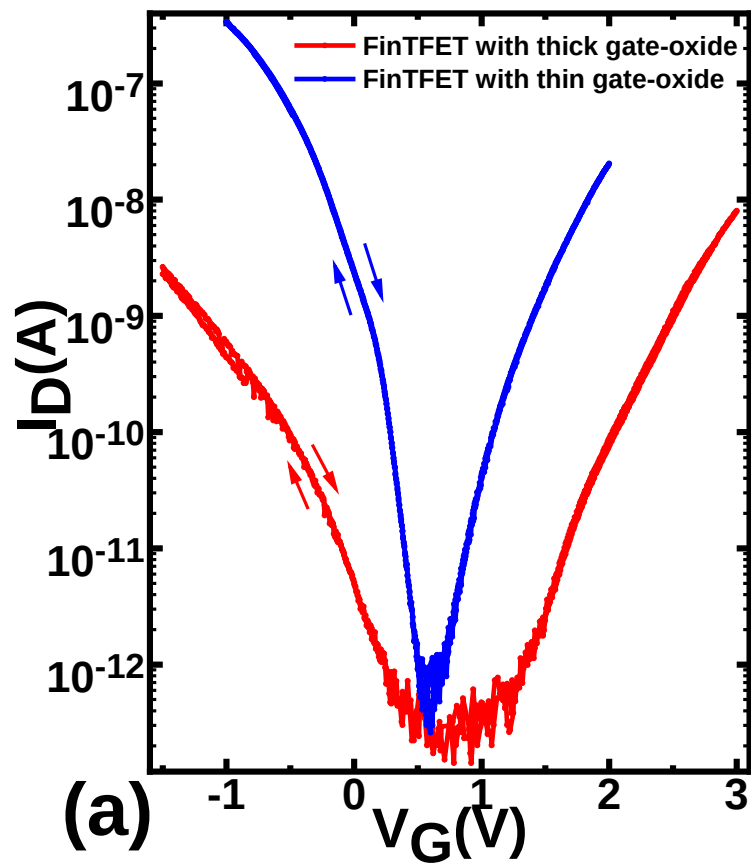
Tunnel FinFET (FinTFET)
fabricated on foundry platform
of 14 nm technology node



FinTFET layout with unsymmetrical doping
of drain and source for ambipolarity.

I_D - V_D curves of the Fe-FinTFET





FeTFET-based AND-type MAC with XOR-cipher

AND-based MAC with XOR-cipher

$$y_j = \sum_i^n x_i \cdot w_{i,j}$$

$$= \sum_i^n x_i \cdot (w_{e,i,j} \oplus \text{Key}_i)$$

X	"0"	"0"	"1"	"1"
W	"0"	"1"	"0"	"1"
$X \cdot W$	"1"	"0"	"0"	"1"

$$= \sum_i^n x_i \cdot (\overline{w_{e,i,j}} \cdot \text{Key}_i + w_{e,i,j} \cdot \overline{\text{Key}_i})$$

$$= \sum_i^n (x_i \cdot \text{Key}_i) \cdot \overline{w_{e,i,j}} + \sum_i^n (x_i \cdot \overline{\text{Key}_i}) \cdot w_{e,i,j}$$

Binomial operation in each W_e cell

